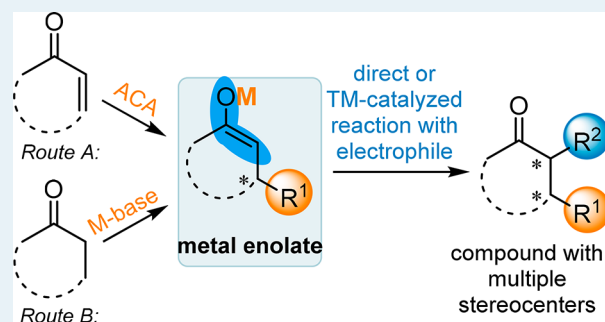


Asymmetric Transition-Metal Catalysis in the Formation and Functionalization of Metal Enolates

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ABSTRACT: This review comprises two major parts, which have chiral metal enolates as their focus. In the first part, we discuss transition-metal-catalyzed conjugate additions leading to chiral enolates, which are then subsequently functionalized by reactions with electrophiles, both further catalyzed or uncatalyzed. The second part deals with base-generated enolates and their use in asymmetric transition-metal-catalyzed processes.



KEYWORDS: asymmetric catalysis, enolate, transition metal, conjugate addition, cross-coupling, domino reaction

INTRODUCTION

Metal enolates and their reactivity are the cornerstones of the classical organic chemistry repertoire for the construction of new molecules. Apart from prototypical lithium enolates, excellently covered in a classical review by Seebach,¹ there can be a multitude of metals used to generate enolates. Deprotonation of corresponding carbonyl compounds and conjugate addition (CA) to α,β -unsaturated carbonyl compounds are two of the most common ways for the formation of enolates, but other less common methods are also possible. This review comprises two major parts, which have chiral metal enolates in their focus. In the first part, we discuss transition-metal-catalyzed CAs leading to chiral enolates, which are then subsequently functionalized by reactions with electrophiles, both further catalyzed or uncatalyzed. The second part deals with base-generated enolates and their use in asymmetric transition-metal-catalyzed processes.

Generation of metal enolates by catalytic CA of organometallic reagents and their subsequent electrophilic trapping in an achiral way has been known since the 1980s. This review does not deal with conjugate additions of stabilized nucleophiles, which has recently been reviewed by Feng.² In the late 1990s, enantioselective variations were developed, especially with zinc enolates. These early developments were highlighted in a minireview by Ma and Guo,³ and also by our group.⁴ In the field of CA, copper is the metal of choice used for catalysis more than any other metal. The copper-catalyzed CA of organometallic reagents to Michael acceptors has become a powerful method for stereoselective construction of new C–C bonds.^{5–11} The methodology has improved over the years, and currently it provides a wide variety of products in high yields and enantioselectivities, including complex natural products.¹²

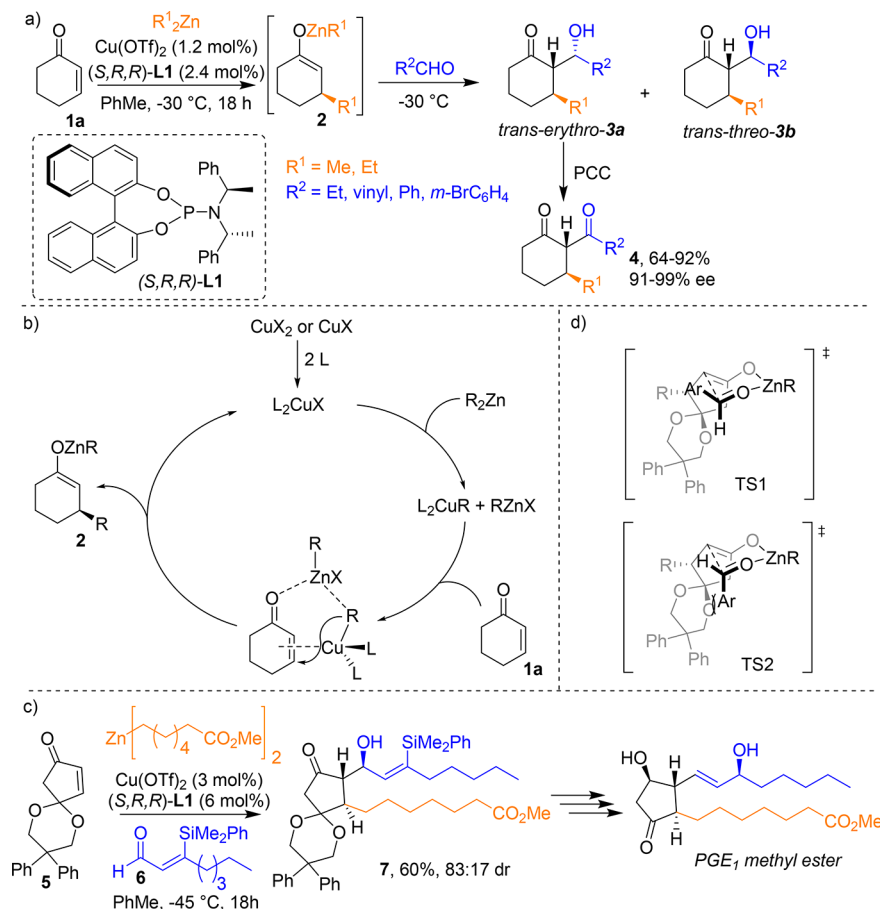
Among the tandem conjugate addition–trapping reactions, where more than one new C–C bond is formed, the use of zinc enolates is the most widely occurring one. Conjugate addition of organozinc reagents is a well-established methodology. With a variety of well-performing ligands available, it is easy to find one suitable for a chosen substrate. Unfortunately, despite decades of research in this field, the scope and commercial availability of organozinc reagents are very limited. The methodology is mostly suited for adding simple alkyl chains, since only one of the groups of the dialkylzinc reagents ends up built into the molecule. Hence, it is not suitable for adding more valuable alkyl chains. Grignard reagents are the second most useful reagents, with a better atom economy in comparison to organozinc reagents. The reactivity of the magnesium enolates is slightly higher than that of zinc enolates. In some cases, unreactive aggregates are formed, and the use of additives is required to break them apart. Zirconium reagents are advantageous regarding the use of noncryogenic reaction conditions. Their added benefit is that they do not react as violently with water as other organometallic reagents, which makes their handling easier. Another advantage is the broader scope of the possible alkyl groups which can be added, with a higher tolerance for functional groups. Aluminum enolates are among the least reactive in tandem reactions and are usually outperformed by zinc or magnesium enolates. Silyl enol ethers are featured in both parts of the review. They are beneficial, since they do not form aggregates and are well-defined. On the other hand, fluoride additives are often required to release the enolate, since they

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Scheme 1. (a) The First Example of a Tandem Asymmetric Aldol Reaction, (b) Catalytic Cycle for Cu-Catalyzed Conjugate Addition of Dialkylzinc Reagents to Enones; (c) Synthesis of (–)-Prostaglandin E₁ Methyl Ester by a Tandem Aldol Reaction, and (d) Zimmerman–Traxler Transition State Models for Aldol Reaction of Chiral Zn Enolate Generated from Enone 5



are somewhat less reactive. Base-generated enolates are used mainly for α -aryl and α -allylation reactions. The use of the specific alkali metal is reaction-dependent, but in most cases more options are possible, with sodium enolates being the most abundant ones.

Reactive enolates can also be generated in situ using a strong, in most cases inorganic base and used in a subsequent reaction with an aryl halide catalyzed by a chiral metal–ligand complex. These asymmetric cross-coupling reactions represent a useful methodology to create new stereogenic centers on challenging substrates.¹³ Particularly, α -aryl carbonyl compounds are of great interest in medicinal chemistry and the synthesis of natural products. However, limited methods for their preparation are available, as asymmetric organocatalytic methods are often used for α -arylations of aldehydes.^{14–16} In those cases, where organocatalytic methods cannot be implemented, enantioselective cross-coupling of enolates can be used as a complementary method.

■ ENOLATES GENERATED BY THE CATALYTIC CONJUGATE ADDITION OF ORGANOMETALLIC REAGENTS

Zn Enolates. Alexakis and Feringa realized the first examples of trapping reactions, the first one being an aldol reaction in 1997.¹⁷ This was a groundbreaking work, since it was the first time that total stereocontrol in the CA was achieved. The new type of phosphoramidite ligand **L1** was

introduced, which helped reach high enantioselectivities of the CA (up to 99% ee). Eventually, this ligand (S,R,R) -**L1** proved to be one of the most versatile and robust in the CA chemistry and is often the ligand of choice for many substrates. A tandem aldol reaction was performed on the zinc enolate **2**, giving a mixture of *trans*-erythro-**3a** and *trans*-threo-**3b** products in a combined 85% yield (Scheme 1a). After oxidation with PCC, the corresponding diketone **4** was obtained in 81% yield with high enantiomeric purity (93% ee).

Zinc enolates **2** usually require more reactive alkylating reagents, because they are rather unreactive nucleophiles.^{18,19} To compensate for their low reactivity, palladium or iron catalysis is often necessary,^{20–26} because the metal allyl complexes formed in situ are more reactive electrophiles. A possible pathway was proposed by the authors on the basis of the generally accepted mechanism for the 1,4-additions of zinc reagents to Michael acceptors (Scheme 1b).^{27,28}

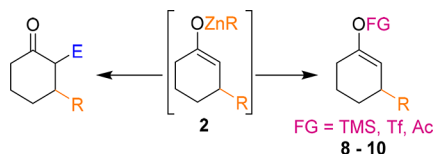
A critical application of zinc enolate trapping was the synthesis of (–)-prostaglandin E₁ methyl ester, which was prepared by a catalytic multicomponent reaction using only achiral starting materials **5** and **6** (Scheme 1c).²⁹ The enantioselective 1,4-addition was performed using 3 mol % of $Cu(OTf)_2$ and 6 mol % of phosphoramidite ligand (S,R,R) -**L1**. Enantioselectivities of the 1,4-addition were up to 94%, and for the tandem 1,4-addition–aldol reaction 97% ee was achieved. The major diastereomer of the aldol reaction can be invoked from the standard Zimmerman–Traxler transition

state model TS1 (Scheme 1d). It is also assumed that the chiral copper ligand complex does not participate in the aldol reaction. The excellent stereoselectivity of the aldol reaction shows that this methodology provides a useful synthetic route to enantiomerically pure multifunctional cyclopentanones.

As aldol reactions in enolate trapping were reviewed in 2006,³ we will not go much further into details. We will only briefly mention the work of Feringa, who used chiral Taddol-based phosphoramidites and bidentate BINOL-derived phosphoramidite ligands for a 1,4-addition and subsequent aldol reaction with benzaldehyde on cyclopentenone.^{30,31} Alexakis also used phosphoramidites to trap zinc enolates **2** with acetals, ketals, and ortho esters, which upon treatment with Lewis acids such as $\text{BF}_3 \cdot \text{Et}_2\text{O}$ afforded corresponding carbonyl compounds in situ.³²

The chiral zinc enolates **2** can be transformed in situ to give more functionalized products.³³ According to the softness or hardness of the added electrophile, the reaction can occur primarily on the oxygen or the C2 carbon atom. Since there are several possibilities to trap the enolate, the synthetic value of this method is significant. Functionalization on the oxygen atom includes transmetalation of **2** to silyl enol ethers **8** and the formation of triflates **9** and acetates **10** (Scheme 2). Carbon–carbon bonds can be formed by allylic alkylation (AA) or reactions with other C electrophiles.^{26,34–36}

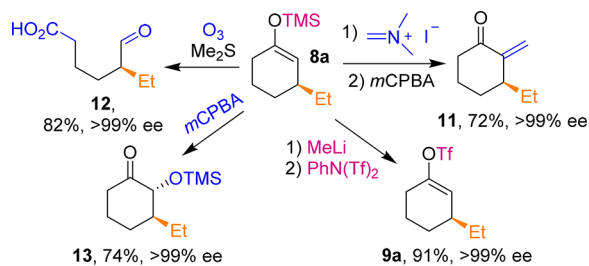
Scheme 2. Trapping of Zinc Enolates **2** To Form Tandem Products or Change of Functional Group on Oxygen



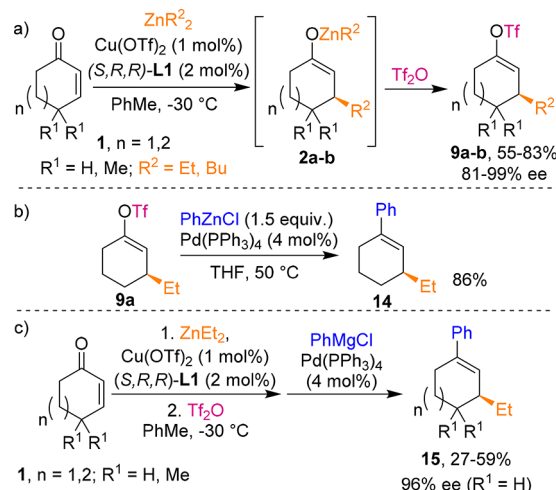
Alexakis reported a variety of products resulting from trapping of zinc enolates **2**. First, the zinc enolate **2** was transformed to silyl enol ether **8a** by $\text{TMSCl}/\text{Et}_3\text{N}$ or TMSOTf .³⁴ Then it was transformed into other chiral synthons. Trapping by Eschenmoser's salt gave the product **11** with a methylene group in the α position. Transmetalation with MeLi and the addition of PhNTf_2 gave the corresponding triflate **9a**. Silyl enol ethers **8a** can also be oxidized by ozone to the corresponding oxocarboxylic acid **12** or by *m*CPBA to afford silyloxy ketones **13** (Scheme 3).

Feringa took the tandem reaction even further and added an in situ cross-coupling step of the triflate **9a** with an additional organometallic reagent.³⁷ CA of the organozinc reagent to a cyclic enone **1**, catalyzed by the Cu/phosphoramidite complex (*S,R,R*)-L1, afforded the chiral zinc enolate **2** (Scheme 4a).

Scheme 3. Various Possibilities of Transforming the Intermediate Silyl Enol Ether **8a** into Different Products



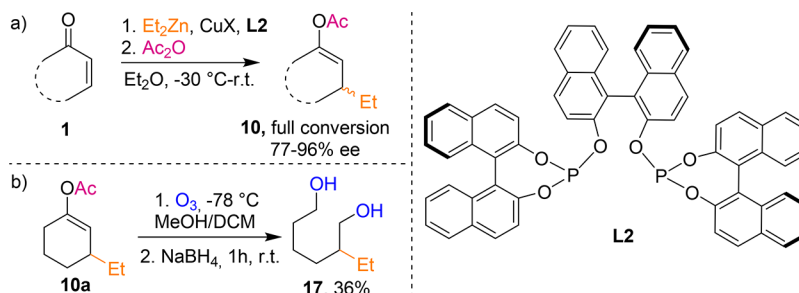
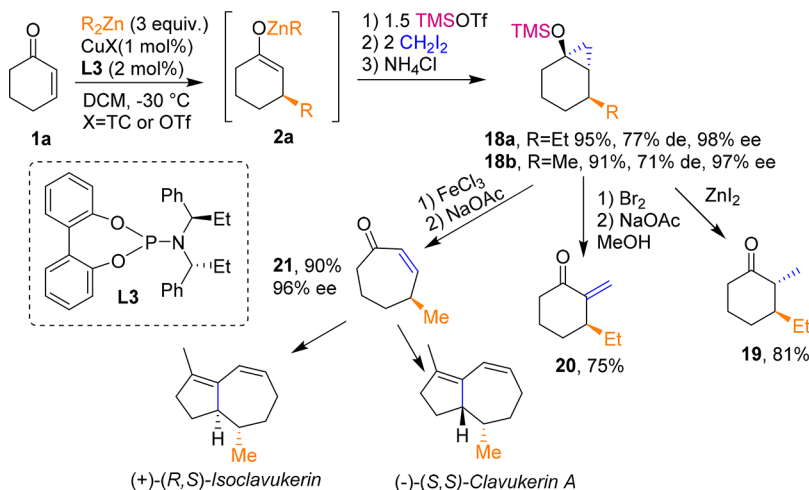
Scheme 4. In Situ Transformation of Zinc Enolates **2** into (a) Alkenyl Triflates **9** and Their Subsequent Use in (b) Cross-Coupling after Isolation or (c) Direct Reaction



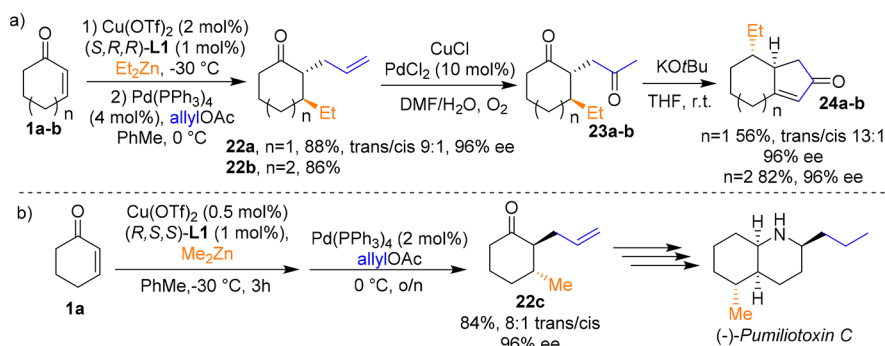
Zinc enolates **2a,b** were then transformed to the corresponding alkenyl triflates **9a,b**. The triflates **9a,b** were obtained in 65–83% yields and with excellent enantioselectivities (94–99% ee) when diethylzinc was used. With dibutylzinc the yields were lower, but the enantiomeric purities of the products remained high (81–95% ee). The alkenyl triflates **9a,b** could be used in a cross-coupling with PhZnCl after isolation (Scheme 4b) or in situ with PhMgCl (Scheme 4c).

Alexakis also reported trapping of enantiomerically enriched Zn (**2**) and Al enolates (**16**) using acetic anhydride to give the corresponding enol acetates **10**, which are more stable than silyl enol ethers **8** (Scheme 5a).³⁵ For each enone, the corresponding enol acetate **10** was formed with complete conversion and high enantiomeric purity up to 96% ee. A similar formation of enol acetates **10** was also described with acyclic enones **1**, both aliphatic and aromatic. The reaction proceeded with almost full conversion (>98%) and enantioselectivities in the range of 60–80% ee. Interestingly, by interchanging organozinc and organoaluminum species, opposite stereoisomers on the double bond could be achieved. The authors speculated that the origin of this chemoselectivity was in the different Lewis acidity of the organometallic species. By using organoaluminum species, the Lewis acidity of the organometallic species increases, and a stronger interaction should have been observed between the oxygen atom of the carbonyl moiety and the metal. This effect should have favored the *s-trans* conformation of the enone, but the opposite was observed—the (*Z*)-enol acetate **10** was mainly formed. The stability of the enol acetate **10** was also determined. The reaction on the model substrate was quenched at 90% conversion, and an *E:Z* ratio of 43:57 was measured. In a parallel experiment, the reaction mixture was stirred at room temperature overnight to reach thermodynamic equilibrium, and an *E:Z* ratio of 52:48 was measured. This result confirmed that the equilibration was very slow and that the (*Z*)-enol acetate **10** was the kinetic product of the reaction. The obtained enol acetates **10a** afforded diols **17** after reductive ozonolysis (Scheme 5b).

Scheme 5. (a) CA and Subsequent Trapping into Enol Acetate 10 and (b) Ozonolysis of Enol Acetate 10a

Scheme 6. Tandem Cyclopropanation of Silyl Enol Ether **8** and Subsequent Ring Cleavage

Scheme 7. (a) The First Tandem CA–Allylic Alkylation and (b) Synthesis of (–)-Pumiliotoxin C



The double bond of the zinc enolate **2a**, after its transformation to the silyl enol ether **8**, was used as a starting point for cyclopropanation with CH_2I_2 (Scheme 6).¹⁸ The cyclopropane ring of **18** was then selectively cleaved in different ways. Cleavage by a Lewis acid provided the corresponding α -methyl- β -alkyl ketone **19**. A bromination–dehydrobromination sequence leads to the α -methylene- β -alkyl ketone **20**. The radical-induced opening of the cyclopropane ring was used to open the ring to give the enone **21** with a γ -alkyl expanded chain. The ring-opening procedure did not affect the enantiomeric purity of the products, which makes this methodology useful for the synthesis of valuable chiral building blocks. It was later applied to the synthesis of (–)-(S,S)-clavukerin A and (+)-(R,S)-isoclavukerin, bioactive sesquiterpenes isolated from the Okinawan soft coral *Clavularia koellikeri*.

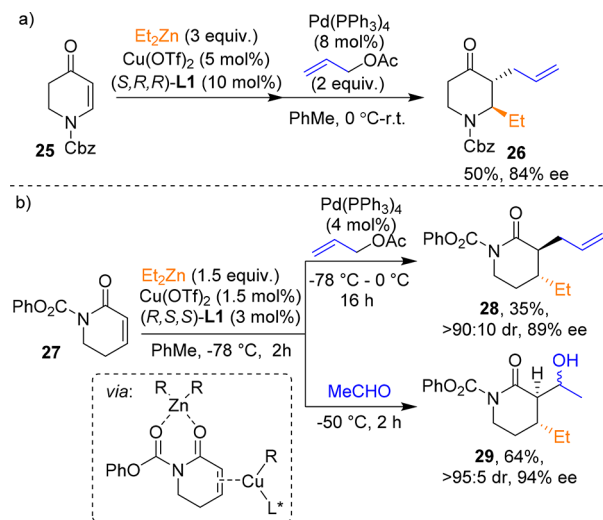
The Feringa group published the first tandem CA/AA using zinc enolates **2** in 1999 (Scheme 7a).²¹ This methodology was used to create 5,6- and 5,7-annulated systems. Enolate **2** was formed in a typical way by Cu phosphoramidite catalyzed CA of dialkylzinc reagent to enones **1a,b**. Zinc enolate **2** was then used in a one-pot Pd-catalyzed allylation reaction with allyl acetate, leading to the corresponding 2-allyl-3-ethylcyclohexanone **22a** in 88% yield, 9:1 trans:cis ratio, and 96% ee. A subsequent Pd-catalyzed Wacker oxidation and aldol cyclization afforded the cyclization products **24a,b**.

Feringa applied a tandem allylic alkylation in the synthesis of pumiliotoxin C (Scheme 7b), a neurotoxin isolated from poison dart frogs, which has attracted the attention of chemists because of its perhydroquinoline moiety with four stereogenic centers.²⁴ The critical step of the asymmetric synthesis of pumiliotoxin C was a tandem CA of dimethylzinc to enone **1a** followed by Pd-catalyzed AA. The tandem product **22c** was

obtained in 84% yield, 8:1 *trans:cis* ratio, and high enantiomeric purity of 96% ee.

Feringa's group reported CA and the trapping of the zinc enolate **2** on less reactive substrates, such as *N*-protected 2,3-dehydro-4-piperidones **25** (Scheme 8a).³⁶ The allylated

Scheme 8. (a) CA and Trapping of the Zinc Enolate **2** on Piperidone **25** and (b) Trapping Reactions of Zinc Enolates **2** Derived by CA of *N*-Protected Lactams **27**



product **26** was obtained as a single *trans* diastereomer, in 50% yield and 84% ee. Unsaturated lactams **27** with carbamate protecting groups were also reported to undergo CA and trapping of the corresponding zinc enolate **2** by allylic alkylation (Scheme 8b).³⁸ *N*-Carbonyl lactams show different reactivity in comparison to the *N*-alkyl-protected lactams, due to the greater electron-withdrawing ability of the carbonyl group. Consequently, the β -carbon becomes more electrophilic. A reasonable assumption is also that the alkyl metallic

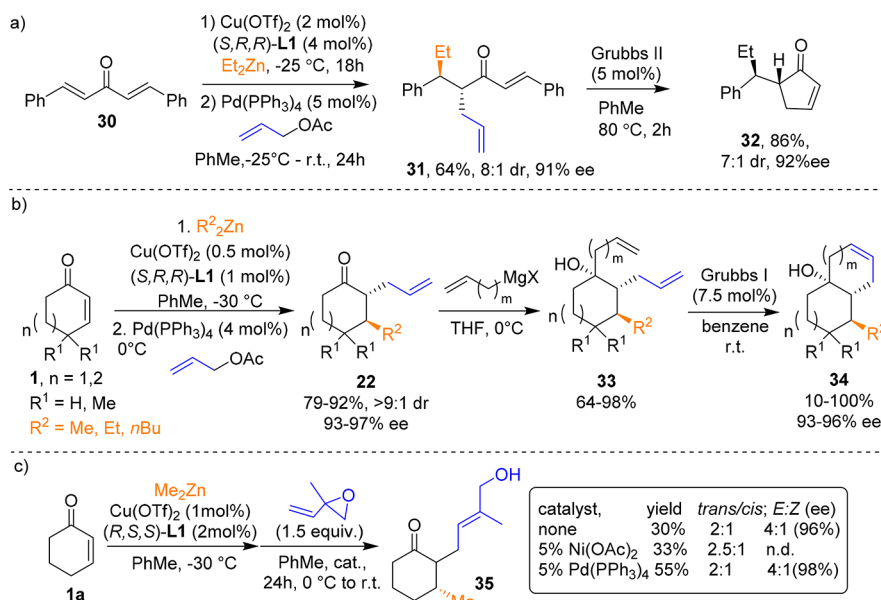
species coordinates the two carbonyl groups, and the chelation might activate the double bond even further. The zinc enolate **2** reacted with allyl bromide or allyl acetate in the presence of a Pd catalyst. The resulting product **28** was isolated in only 25–35% yield, albeit with a high enantiomeric purity of 89% ee. The addition of acetaldehyde to the zinc enolate **2** gave an inseparable mixture of aldols **29** with 94% ee and 64% yield.

A tandem CA–allylic alkylation can also be performed on dibenzylidene acetone **30** using the typical Cu/(*S,R,R*)-L1 catalytic system (Scheme 9a).³⁹ Pd catalysis was required for the AA with allyl acetate. The tandem product **31** was obtained in 64% yield, 8:1 dr, and 91% ee. Taking advantage of the two double bonds in the tandem product, the authors performed a Ru-catalyzed alkene metathesis, which afforded the enantioselectively enriched α,β -unsaturated cyclopentenone **32** in 86% yield, 7:1 dr, and 92% ee.

Ring-closing metathesis on tandem alkylation products **22** was used to prepare bicyclic compounds **34** (Scheme 9b).²³ The typical combination of phosphoramidite (*S,R,R*)-L1 and Cu(OTf)₂ was used in the CA step. The trapping by allyl acetate and a catalytic amount of Pd(PPh₃)₄ gave the allylated products **22** in good yields of 79–92% and diastereoselectivities ranging from 5.3:1 up to 196:1. The enantioselectivities were 93% ee or higher. By treatment of the tandem products **22** with a Grignard reagent with a terminal double bond, the products of 1,2-addition **33** were obtained, which were ideally suited for the ring-closing metathesis. A reasonable selectivity was observed for the addition of the Grignard reagent *trans* to the allyl group, leading to the all-*trans* isomer as the major product. Products with [6,6], [7,6], [8,6], and [6,7] carbocyclic skeletons were prepared (Scheme 9b).

Alexakis developed a method for trapping the zinc enolate **2** by isoprene monoxide (Scheme 9c).²⁵ The zinc enolate **2** yielded the tandem product **35** even without an additional metal catalyst in 30% yield and 96% ee. The use of a Ni catalyst did not affect the yield, but the addition of a Pd catalyst improved the yield to 55%, and the major diastereomer

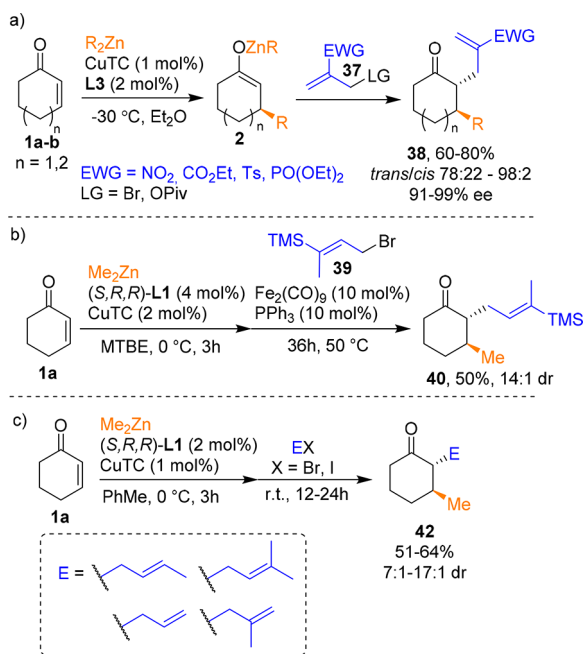
Scheme 9. (a) Tandem CA–Allylic Alkylation on Dibenzylideneacetone (**30**), Followed by Ru-Catalyzed Metathesis, (b) CA and Allylic Alkylation Sequence Followed by 1,2-Grignard Addition and Ring-Closing Metathesis, and (c) Trapping of the Zinc Enolate **2** by Isoprene Monoxide



was obtained in high enantiomeric purity (98% ee). Aluminum enolates **16** could also be used, with comparable results, but this was not the case for magnesium enolates **36**, from which no product was obtained. The *E*:*Z* ratio can shift toward the exclusive formation of *E* isomer by Swern oxidation of the allylic alcohol to an aldehyde. Not surprisingly, butadiene and isoprene monoxide gave the best results. The reaction proceeded with moderate *E*:*Z* and *trans*:*cis* selectivities (typically 2:1 and 3:1, respectively). Applicability of the newly developed methodology was demonstrated in the synthesis of a precursor of the anticancer agent clavularin B.

Alexakis described trapping of zinc enolate **2** by activated electrophiles **37** without the need for another metal (Scheme 10a).²⁶ These electrophiles **37** had an electron-withdrawing

Scheme 10. (a) Trapping of **2** by Activated Allylating Reagents **37**, (b) Trapping of **2** by Stork–Jung Electrophile **39** under Iron Catalysis, and (c) Trapping of Zinc Enolates **2** without the Additional Catalyst in PhMe



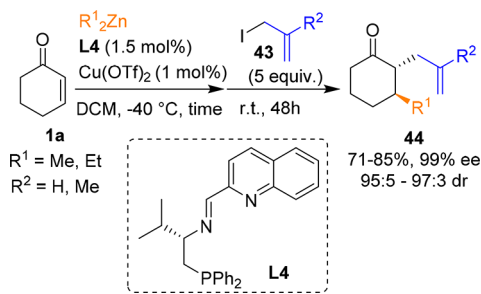
group in the vinylic position and a leaving group in the allylic position. The reaction is considered as an allylic substitution, even though it probably proceeds by a CA–elimination mechanism. The undesired reaction between the residual diethyl zinc and the electrophile was slowed down by lowering the temperature. The addition of a catalytic amount of $\text{Pd}(\text{PPh}_3)_4$ increased the yields of trapping products **38**. A higher yield of the trapping product **38** can also be achieved simply by adding a larger amount of the electrophile. When CuTC was used instead of $\text{Cu}(\text{OTf})_2$, high enantioselectivities of the trapping products **38** were obtained (91–99% ee). The enantiodetermining step is believed to be the conjugate addition, and the allylation follows in a typical fashion from the least sterically hindered side of the enolate, resulting in the *trans* product.

Cook reported an interesting allylation of zinc enolates **2** using an iron catalyst.⁴⁰ He also used Stork–Jung electrophile **39** for the alkylation of zinc enolates **2** (Scheme 10b).⁴¹ Stork–Jung acetate did not give any product **40**, but iodide and bromide gave products under additional iron catalysis in

moderate yields and high diastereoselectivities. Interestingly, diastereoselectivities were higher in comparison to the thermodynamic preference for the *trans* isomer, which was determined to be 2.4:1 *trans*:*cis* by base-catalyzed epimerization studies. Surprisingly, palladium catalysis gave inferior results in this case. The authors have also observed that ethereal solvents provided low yields of the tandem products **40** (less than 40%), but the use of toluene increased the yield to 80% (Scheme 10c). A possible explanation of this effect was that in toluene the more reactive dimeric structure is dominant. In ethereal solvents, on the other hand, an equilibrium between the dimeric structure and less reactive aggregates is established. In lithium enolates **41**, aggregate states strongly affect their reactivity, and probably something similar happens with zinc enolates **2** as well.⁴² Diethylzinc reacted less efficiently in comparison to dimethylzinc in the allylation reaction, probably due to a side reaction with the electrophile. As in many other cases, the reaction with cyclopentenone failed to give any product under the standard reaction conditions.^{17,21,43–49}

Hayashi and co-workers used allyl iodide **43** to trap zinc enolates **2** (Scheme 11).⁵⁰ They observed that the amount of

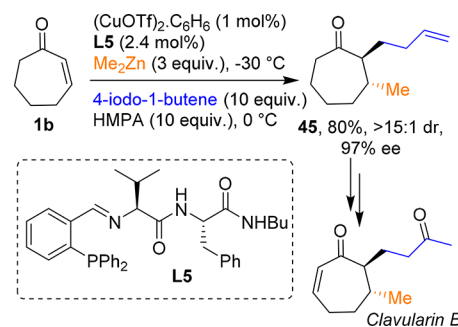
Scheme 11. Hayashi's Trapping of Enolates **2** by Allyl Iodide **43**

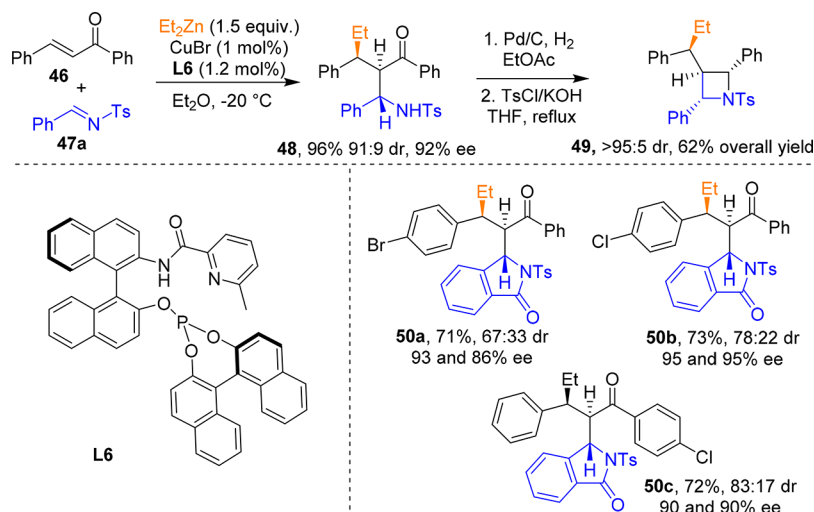
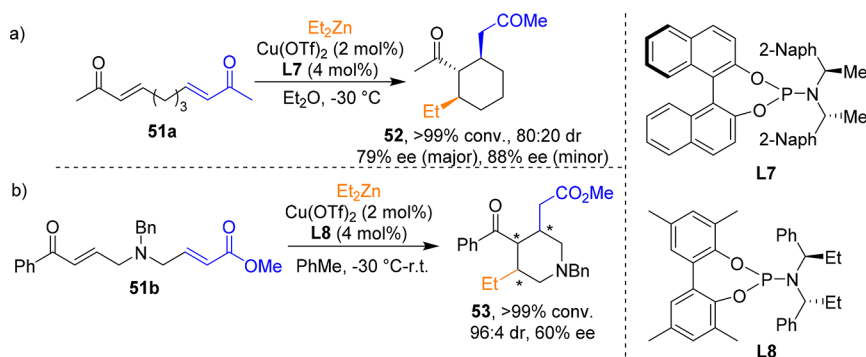
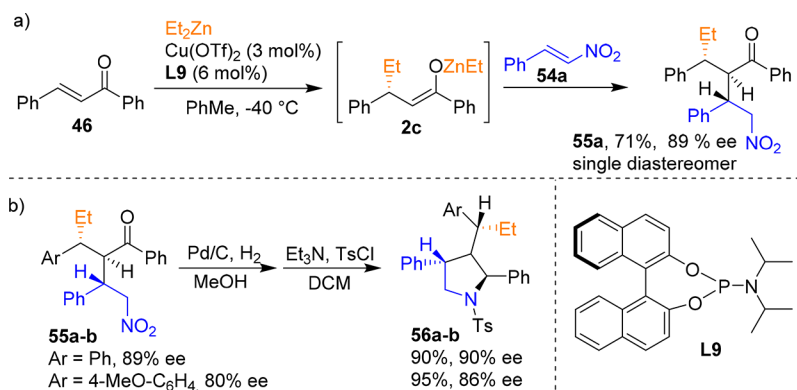


dialkylzinc had a crucial role in the success of the reaction. Excess dialkylzinc acted as a base and generated the new zinc enolate **2** from the allylated product **44**, which led to a diallylation. For the CA the new *N,N,P*-ligand L4 with a quinoline moiety and $\text{Cu}(\text{OTf})_2$ was used.

Hoveyda used the peptide-based chiral phosphine ligand L5 for the CA of dialkylzinc to enone **1b** and trapped the enolate **2** by adding a large excess of 4-iodo-1-butene (Scheme 12).⁴³ After Pd-catalyzed oxidation and Wacker oxidation the anticancer agent clavularin B was obtained in 42% overall yield.

Scheme 12. CA of Dialkylzinc to Enone **1b** and Trapping by 4-Iodo-1-butene To Give a Precursor for Clavularin B



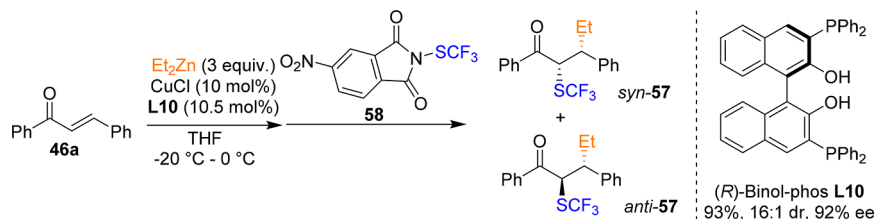
Scheme 13. Tandem Mannich Reaction of Zinc Enolates **2** and *N*-Tosylaldimine **47a**Scheme 14. Spontaneous Intramolecular Cyclization after CA to Acyclic Enones **51a,b**Scheme 15. (a) Trapping of Chiral Zn Enolate **2c** by Nitrostyrene **54a** and (b) Cyclization of Chiral Adducts **56** to Ts-Protected Pyrrolidines **54**

Huang and co-workers developed a highly enantioselective addition of diethylzinc to chalcone-type substrates using binaphthyl ligand **L6** (Scheme 13).⁵¹ The chiral enolate **2** was subsequently trapped by a Mannich reaction with the aldimine **47** to yield chiral β -aminocarbonyl derivatives **48**. The reaction did not proceed with less electrophilic imines such as *N*-arylimine and *N*-alkylimine and required *N*-tosylaldimine **47a**. Notably, only two of the four possible diastereomers were detected. The counterion of the copper salt greatly affected the outcome of the reaction. More coordinating counterions such as CuBr proved to be more suitable for the reaction. The nature of the substituents did not

have a large effect on the outcome of the reaction. The corresponding products **48** were obtained in up to 99% yield, 95:5 dr, and 95% ee. A nosyl (4-nitrobenzenesulfonyl) protecting group on the nitrogen instead of tosyl could also be used. The tandem reaction products **48** were further transformed by lactamization into chiral *N*-tosylphthalimides **50** in good yields and enantioselectivities. The tandem adducts **48** could be transformed into highly substituted azetidines **49** in two steps.

Alexakis published an intramolecular enolate trapping reaction, which resulted in cyclic compounds **52** with multiple stereogenic centers (Scheme 14).⁵² The cyclization step turned

Scheme 16. Trifluoromethylthiolation of Chiral Zinc Enolates 2



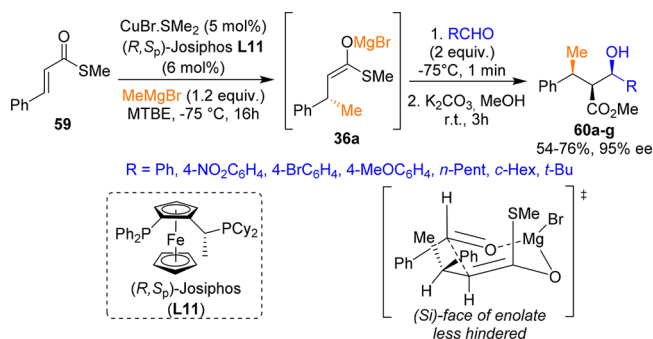
out to be faster than the second CA of the dialkylzinc reagent, since no second CA product was observed even with an excess of organozinc reagent. The diastereoselectivities were comparable with all ligands, and moderate enantioselectivities were obtained. Ligand **L7** gave the best enantioselectivity (79 and 88% ee) for the two isomers when Et_2Zn was used (Scheme 14a) and 88% and 94% ee when Me_2Zn in toluene was used. Chiral piperidine **53** was also prepared, but in this case, the enantioselectivity of the reaction dropped to 60% ee (Scheme 14b).

Liu and co-workers trapped chiral zinc enolates **2** by nitro olefins **54** to prepare precursors **55** for chiral pyrrolidines **56** (Scheme 15).⁵³ Phosphoramidite ligand **L9** gave the best results for the reaction of chalcone **46** with diethylzinc and the trapping of the zinc enolate **2c** by *trans*- β -nitrostyrene **54a** (Scheme 15a). The obtained diastereomer is a result of the *Si* attack on Zn enolate **2**, which is sterically more favorable. The minor diastereomer from the *Re* face approach was not observed in most cases. The reaction conditions tolerated a range of chalcone derivatives, including functional groups such as halogens, OMe, NO_2 , and heterocycles. The chemical yield significantly decreased when cyclohexyl-substituted nitroalkene **54** was used (25% yield, 79% ee), and other aliphatic nitroalkenes were not applicable in this transformation. Increased steric hindrance in the nitro olefin **54** led to lower product **55** yields, albeit with higher enantioselectivity. The transformation of the tandem products **55** into two chiral pyrrolidines **56** was achieved via reduction and *N*-protection (Scheme 15b).

Introduction of fluorinated substituents, such as trifluoromethyl⁵⁴ and difluoroalkyl,⁵⁵ via catalytic processes is an important topic in organic synthesis. The Wang group used chiral zinc enolates **2** to introduce a trifluoromethanethiol group in the α -carbonyl position (Scheme 16).⁵⁶ Many methods for introducing the SCF_3 moiety into organic molecules are already available,^{57–59} but there are only a few for asymmetric trifluoromethanethiolation.⁶⁰ The major challenge with this reaction, especially when the asymmetric transformation is required, is possible bistrifluoromethanethiolation if two α -hydrogens are present. Binol-phos **L10** in combination with CuCl afforded the product **57** in a reasonable yield of 67% and an enantiomeric purity of 93% ee. The screening of SCF_3 reagents revealed that phthalimide reagent **58** gave the best results in terms of yield, diastereoselectivity, and enantioselectivity. Under optimized conditions, the product **57** was isolated in 93% yield, 16:1 dr, and with 92% ee.

Magnesium Enolates. There are various ways to prepare Mg enolates **36**, such as (a) reductive metal insertion into carbon–halogen bonds, (b) halogen–metal exchange, (c) transmetalation, (d) CA, (e) metalation, and (f) miscellaneous methods (e.g., addition to carbon–carbon multiple bonds).^{33,61} Arguably, the most effective approach for the

preparation of Mg enolate **36** intermediates is a CA of Grignard reagents to α,β -unsaturated carbonyl compounds.^{33,62,63} However, use of Mg enolates **36** in asymmetric domino reactions has seldom been described. Feringa showed that Mg enolate **36a** formed by asymmetric Cu-catalyzed CA to α,β -unsaturated thioester **59** could effectively engage in aldol reaction with aldehydes (Scheme 17).⁶⁴ High diaster-

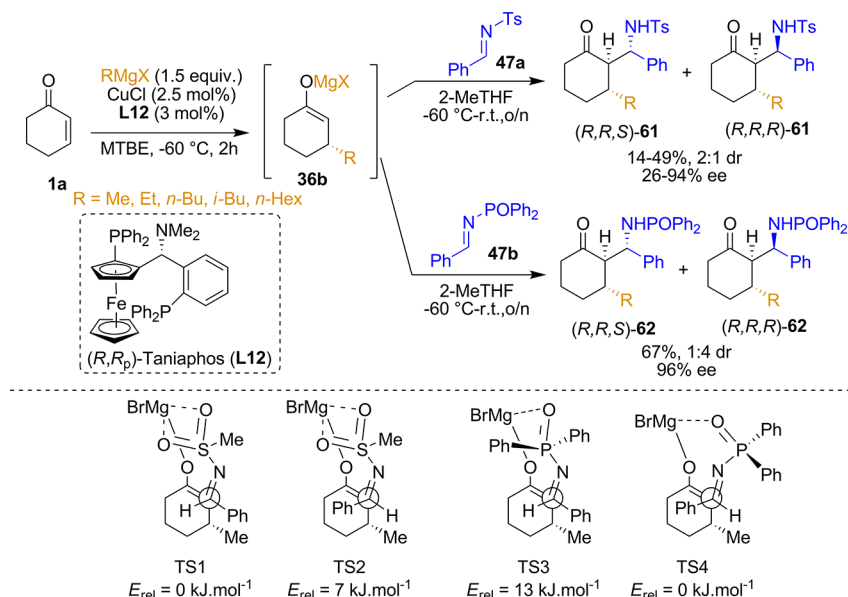
Scheme 17. Trapping of Mg Enolate **36a** by Aliphatic and Aromatic Aldehydes Catalyzed by (R,S_P) -Josiphos **L11**

eselectivity and enantioselectivity of the reaction were explained through a chairlike Zimmerman–Traxler transition state (TS) model. Minimization of the inherent *syn*-pentane interaction in TS would favorize *Si* face attack on enolate leading to the *syn,syn* diastereomer. This transformation tolerates a broad spectrum of aromatic and aliphatic aldehydes. It was also successfully used in the total synthesis of (–)-phaseolinic acid and prodigiosin alkaloids, where the formed Mg enolate **36a** was trapped with chlorosilane.⁶⁵

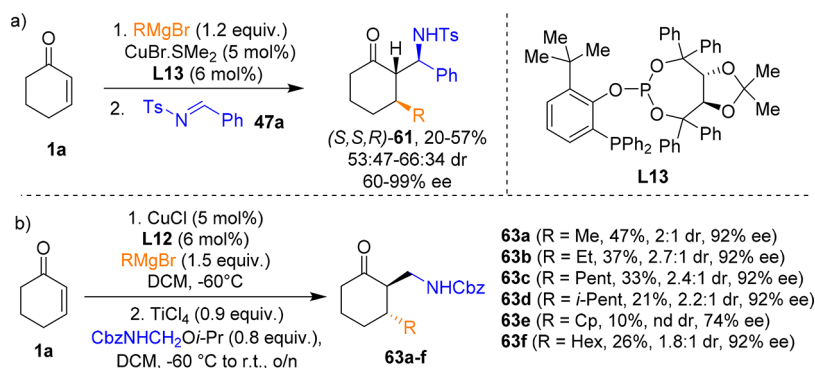
Our group described an enantioselective one-pot CA of Grignard reagents followed by the Mannich reaction (Scheme 18).⁶⁶ Asymmetric CA of different organomagnesium halides to cyclohex-2-enone **1a** produced chiral enolates **36b**, which then reacted with *N*-tosylaldimine **47a**. This reaction led to the synthesis of β -amino carbonyl compounds **61** in good yields and with high enantiomeric excess. As the diastereoselectivity of the Mannich reaction was poor, we have also tested different *N*-protecting groups. For most of the tested protecting groups (Ts, Ms, Bz, and Boc) the diastereoselectivity did not change dramatically. Calculated structures of TS of enolate addition to imines explain that for the sterically less congested imines (protected with Ts, Ms, Boc, and Bz) *Si* face attack is preferred and leads to the formation of (R,R,S) -**61** as the major diastereomer (Scheme 18, TS1 vs TS2). On the other hand, with POPh_2 **47b** dr 1:4 was observed, and imine is attacked from the *Re* face, leading to (R,R,R) -**62** as the major diastereomer (Scheme 18, TS3 vs TS4).⁶⁷

Schmalz's phosphane phosphite ligand **L13** brought about another improvement to the protocol mentioned above. This ligand allowed a broader scope of the tandem CA–Mannich

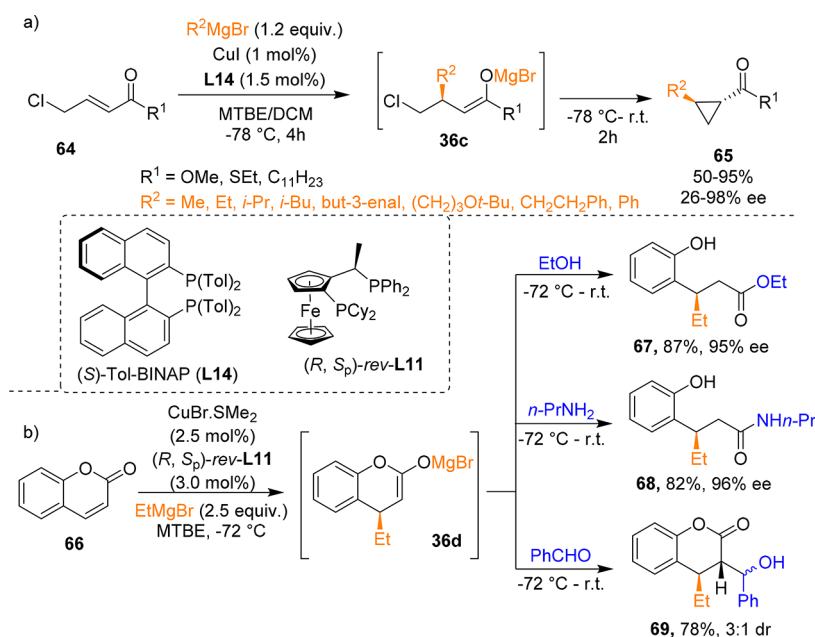
Scheme 18. One-Pot CA of Grignard Reagents Followed by Mannich Reaction



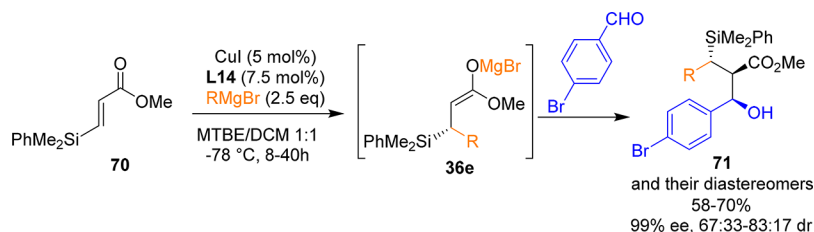
Scheme 19. 1,4-Addition of Grignard Reagents to 1a Followed by Mannich-Type Alkylations



Scheme 20. (a) Intramolecular Trapping of Enolate 36c and (b) Ring Opening and Aldol Reaction of Chiral Enolate 36d



Scheme 21. Aldol Reaction of Chiral Enolate 36e



reaction (Scheme 19a).⁶⁸ In our further studies of Mannich-type reactions of enolates **36b**, we have reported catalytic one-pot CA followed by amidomethylation (Scheme 19b). In situ formed Mg enolates **36b** were trapped with TiCl₄-generated imine, leading to Cbz-protected aminocarbonyl compounds **63** in moderate yields but with high enantiomeric purities (up to 92% ee).⁶⁹

We have also examined ferrocenyl phosphane carbene ligands in Cu-catalyzed 1,4-addition to different types of Michael acceptors.⁷⁰ The Mg enolates **36** were then trapped with imine or benzaldehyde, leading to the formation of products in moderate yields and with high diastereoselectivities.

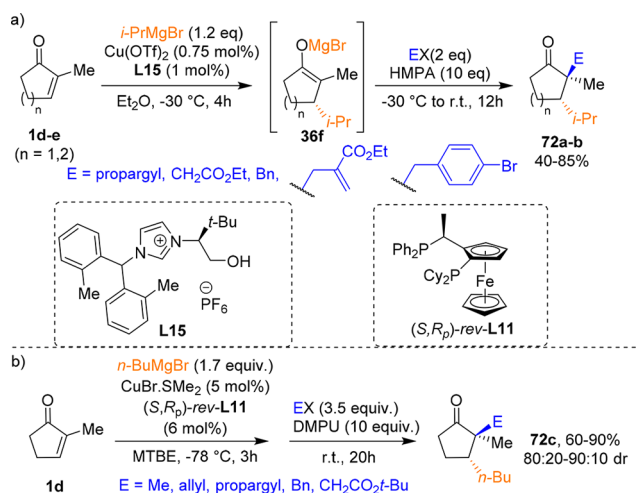
The Feringa laboratory reported tandem Cu-ACA of Grignard reagents to 4-chloro- α,β -unsaturated esters, thioesters, and ketones **64** followed by a ring-closure reaction, yielding *trans*-1,2-disubstituted cyclopropanes **65** (Scheme 20a).⁷¹ They also reported the Cu-catalyzed asymmetric CA of Grignard reagents to coumarins **66** (Scheme 20b).⁷² The chiral Mg enolate **36d** formed was converted in situ to a variety of chiral products (**67–69**). Heteroatomic nucleophiles such as ethanol or amine afforded the ring-opened products **67** and **68** in high yields and enantiomeric purities. Enolate **36d** could be trapped with benzaldehyde to give the corresponding aldol product **69**.

The trapping of in situ formed Mg enolates **36e** by an aldol reaction was described by Loh.⁷³ This reaction afforded a series of highly enantioenriched organosilanes **71** in good yields and with good to moderate diastereomeric purities (Scheme 21).

Alexakis reported the first asymmetric CA of Grignard reagents to substituted cyclic enones **1d,e**.^{74,75} This methodology was then extended to tandem reactions involving CA and enolate trapping, leading to α -substituted cyclopentanones **72a** and hexanones **72b** (Scheme 22a). The reaction afforded products in moderate yields but with excellent enantioselectivity (for cyclohexanone **72b** 80% ee, for cyclopentanone **72a** 84% ee) and diastereoselectivity (87:13 to 99:1 dr). The same group described the one-pot sequence of Michael additions including asymmetric CA and metal enolate trapping with vinyl sulfone and nitroolefin.⁷⁶

The Minnaard group targeted CA to 2-methylcyclopentanone (**1d**) followed by enolate alkylation.⁷⁷ 1,4-Addition of Grignard reagents to this challenging type of Michael acceptor afforded the corresponding enolates, which were then alkylated using 1,3-dimethyltetrahydropyrimidine-2(1*H*)-one (DMPU) as an additive, which is a less toxic alternative to HMPA (Scheme 22b).

Recently, Harutyunyan demonstrated that α,β -unsaturated amides **73** could be activated toward Cu-catalyzed CA with Lewis acids. The corresponding amide enolate **8b** having an

Scheme 22. (a) Tandem ACA/Enolate Trapping Using NHC Ligand L15 and (b) 1,4-Addition of Grignard *n*-BuMgBr to **1b** Followed by Alkylation

adjacent Br substituent then underwent intramolecular alkylation (Scheme 23).⁷⁸

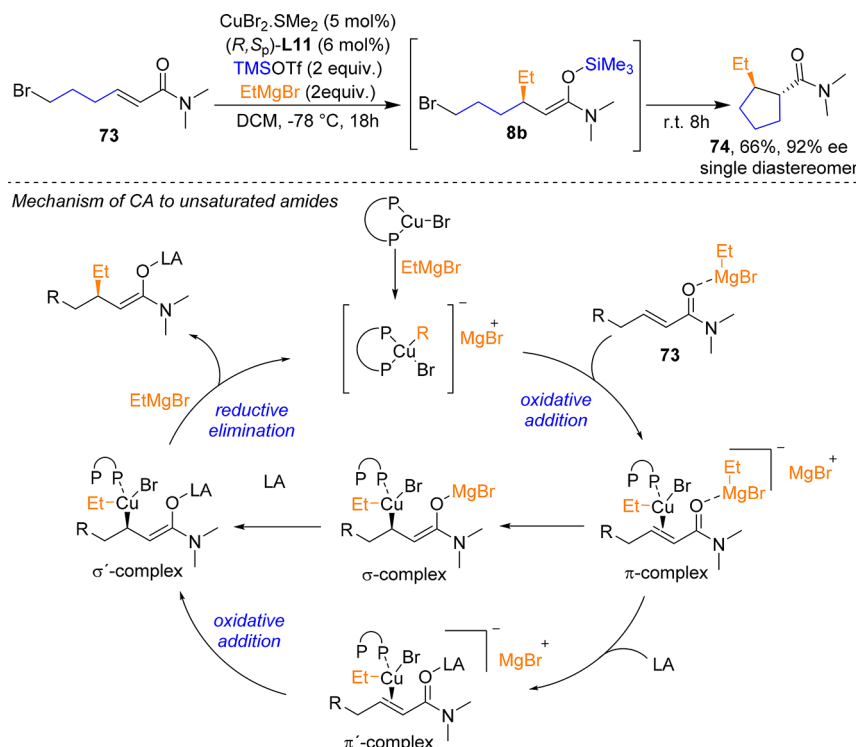
Our group has shown that Mg enolates **36** formed by the addition of Grignard reagents to α,β -unsaturated compounds reacted with naked carbenium ions, such as tropylium **75**, 1,3-benzodithiolium **76**, and dianisylmethylum **77**.⁷⁹ The reaction afforded target products **78–80** in good yields, with excellent enantiomeric purity (up to 96% ee) and high diastereomeric ratio (Scheme 24).

Stereoselective domino CA of Grignard reagents to lactone **81** followed by reaction with activated alkenes **82** was our next target (Scheme 25a).⁸⁰ Ferrocenylphosphane carbene ligand L16 was used to catalyze the CA. The target disubstituted lactones **83** were isolated in medium yields but with high enantiomeric and diastereomeric purities.

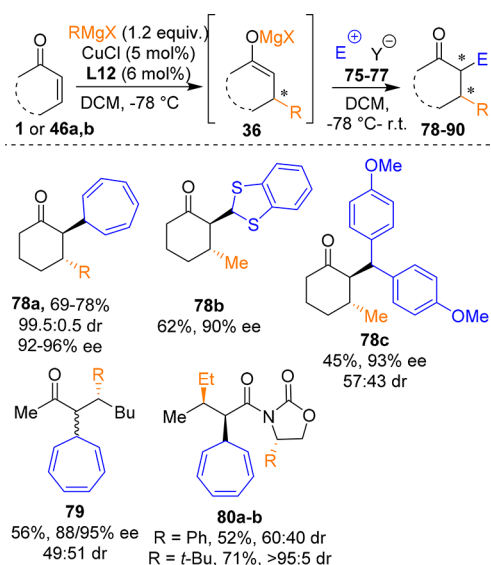
Alexakis has successfully used Michael acceptors such as disulfonyl ethylene **82a** and nitroalkenes **82e** in the enolate trapping reaction for Mg (**36**) and Al enolates (**16**) as well (Scheme 25b).⁷⁶ When 1-alkyl-1-nitroolefins **82e** were used as trapping reagents, an in situ Nef reaction gave the 1,4-diketones **23b** in excellent yields and ees. Mg (**36**) and Zn enolates (**2**) reacted readily, and Al enolates (**16**) reacted well when they were transformed to the corresponding alanates by addition of MeLi. The same trend occurred for the trapping of disulfone **82a**. The products can be transformed into various bicyclic compounds, which represent the structural cores of several sesquiterpenes.

Pronin and co-workers recently utilized Cu-catalyzed CA of Grignard reagents to cyclopentenone **1d** followed by enolate trapping in the total synthesis of (–)-nodulisporic acid C (Scheme 25c).⁸¹ The initially formed Mg enolate **36f** was

Scheme 23. Cu-Catalyzed CA of Grignard Reagents to Unsaturated Amides and Intramolecular Trapping on Enamide 73



Scheme 24. Tandem 1,4-Addition Followed by a Reaction with Naked Carbenium Ions



transformed into the corresponding silyl enol ether **8c**, which was then alkenylated using InBr_3 .

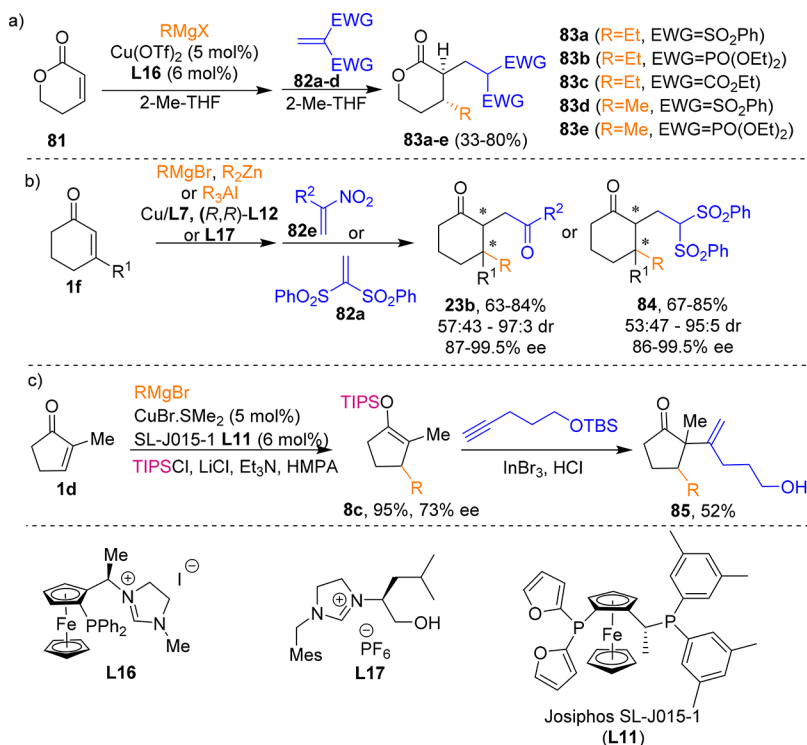
Boron Enolates. Hoveyda and co-workers have studied Cu-catalyzed 1,4-addition of $\text{Ph}(\text{Me})_2\text{Si-B}(\text{pin})$ in the presence of NHC ligands **L18**, followed by an in situ aldol reaction to provide the desired products **86** in high yield and enantiopurity (Scheme 26). The asymmetric incorporation of a silyl group at the β position of the carbonyl compound supposedly enhanced the nucleophilicity of boron-enolate **87a** through hyperconjugation and thus allowed the trapping to occur even on five- and seven-membered cyclic acceptors **1a,b**. There was a lack of products when conventional 1,4-boration

protocols were used. The silyl group can be easily transformed to a hydroxyl group, and thus this method could be complementary to the 1,4-boration methodology.^{82,83} The unsymmetric structure of the NHC-Cu(I) complex influenced the coordination in the transition state. Ligand screening showed that ligands bearing a small *meta* substituent (Me) at Ar^1 and *ortho* substituents (Et) at Ar^2 provided the highest enantioselectivity of the reaction. A large silyl substituent drives the stereoselectivity of the tandem aldol reaction. Later Hoveyda performed broader mechanistic studies concerning this matter; however, they studied more extensively NHC-catalyzed 1,4-additions of borosilane without copper(I) catalyst.⁸⁴

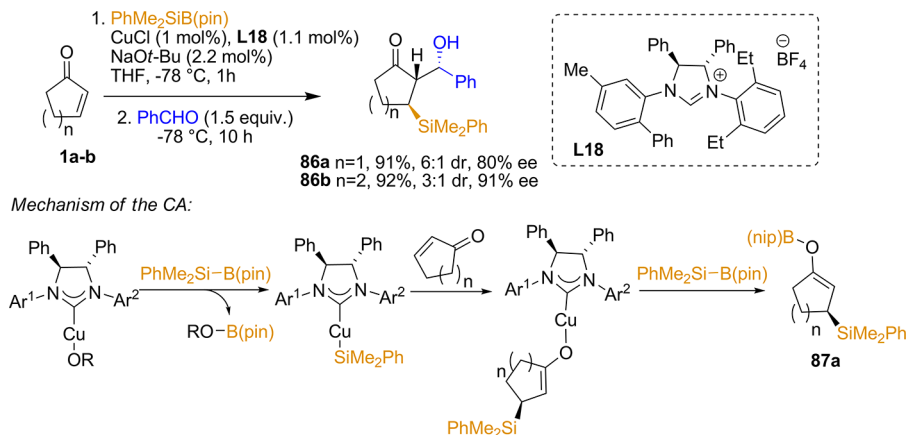
There have been a few reports concerning Cu-catalyzed asymmetric transformations employing B-enolate **87** trapping in a subsequent aldol or Mannich reaction from the Shibasaki laboratory. For the enantioselective Cu-catalyzed intermolecular aldol reaction, allenic ester **88** was chosen as the substrate (Scheme 27). The reaction was initiated by a stoichiometric reduction using $(\text{EtO})_3\text{SiH}$ or $(\text{pin})\text{BH}$. The Cu(I)/(R)-DTBM-SEGPHOS (**L19**) catalytic system was proved to be the best for this reaction in the presence of $(\text{pin})\text{BH}$ as the reducing agent. It provided significantly high selectivity for the γ adducts **89** in a 10:1 ratio for *cis*- γ isomer **89a** in 99% ee. However, the γ/α selectivity was moderate. Importantly, the slow addition of the reducing agent was necessary, because otherwise 1,2-reduction occurred predominantly.⁸⁵

In 2008, Shibasaki reported a synthesis of β -amino esters **92** by performing asymmetric 1,4-reduction of α,β -unsaturated esters **91a** using the catalytic system CuOAc /phosphane ligand with a stoichiometric amount of $(\text{pin})\text{BH}$ or $(\text{EtO})_3\text{SiH}$ as reducing agent. Enolates formed in this way reacted then with various ketimines **47c**. For the diastereoselective version of the reaction $(\text{pin})\text{BH}$ proved to be a better reducing species by providing the desired β -amino esters **92** in diastereoselectiv-

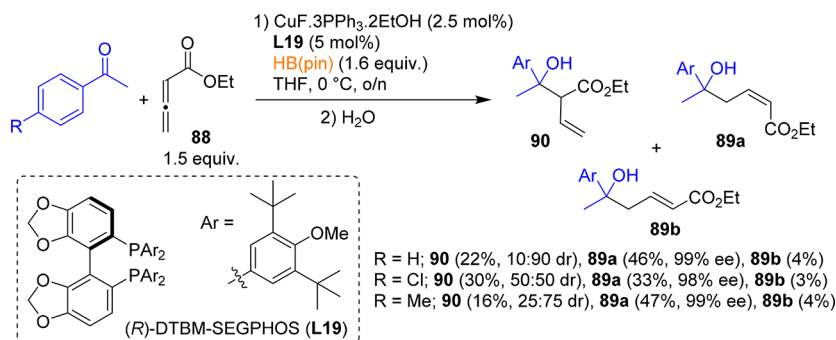
Scheme 25. (a) Tandem 1,4-Addition on Cyclic Lactone **81** and Alkylation by Activated Alkenes **82**, (b) Tandem 1,4-Addition to Enone **1f** Followed by Reaction with Activated Alkenes, and (c) Tandem 1,4-Addition to 2-Methylcyclopentenone, Transmetalation of the Enolate **36** to Silyl Enol Ether **8c**, and Subsequent in-Mediated Alkylation



Scheme 26. Aldol Reaction of Boron Enolate **87a**



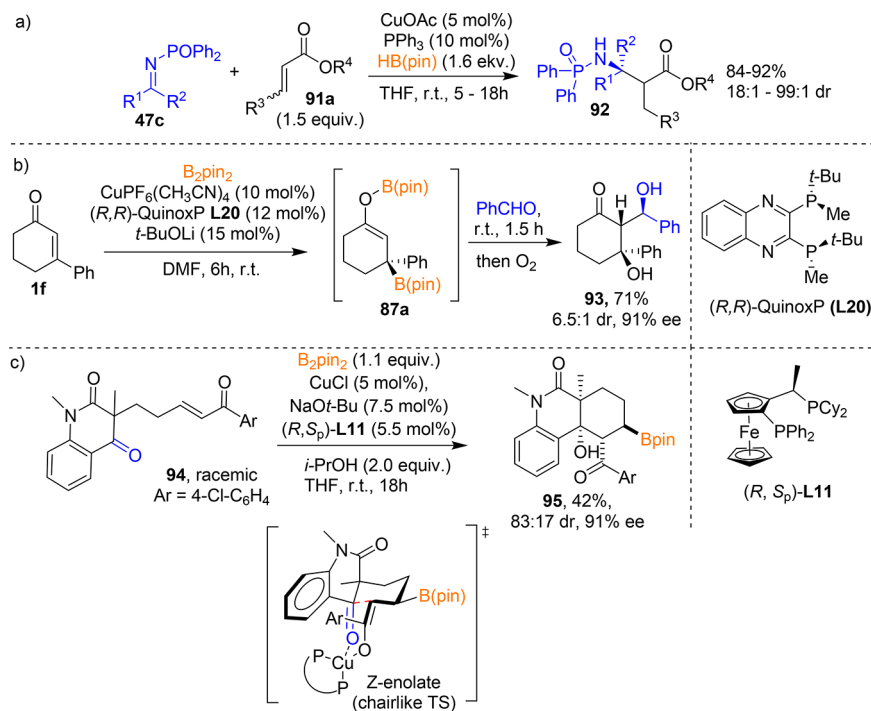
Scheme 27. Aldol Reaction of Boron Enolate **87**, Produced by a Reduction of Allene **88**, and Aryl Ketone



ities up to 99:1 for arylketimines. For the enantioselective version of this reaction, however, the difference in the ee of the

products varied from 25% ee when (pin)BH was used to 68% ee for (EtO)₃SiH. The authors of the study explained this

Scheme 28. (a) Stereoselective Reduction of Ester **91a**, Followed by Mannich Reaction, (b) Trapping of B Enolate **87** by Aldol Reaction with Benzaldehyde, and (c) Borylative Intramolecular Aldol Reaction of **94**



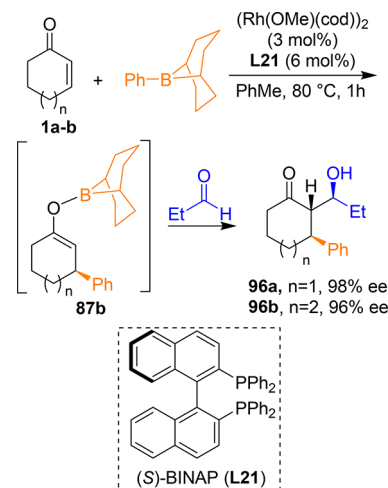
incoherence by the high reactivity of B enolates **87** in comparison to silyl enol ethers **8** (Scheme 28a).⁸⁶ Shibasaki also showed the trapping of B enolate **87** by an aldol reaction with benzaldehyde (Scheme 28b). In this work, Cu-catalyzed 1,4-hydroboration of β -substituted cyclic Michael acceptor **1f** did not require any protic additive and the enolate could react subsequently with electrophiles.⁸⁷

Lam and co-workers have reported a Cu-catalyzed borylative intramolecular aldol reaction.⁸⁸ The desired bicyclic compounds were prepared in good yields and excellent enantioselectivities of up to 99% ee when the CuCl/(*R,S*)-**L11**/NaOt-Bu catalytic system was used. It should be mentioned that the ligand-controlled double-asymmetric process can also explain this enantioenrichment of products excluding the minor enantiomer from cyclization. This reaction showed a good level of enantioinduction on racemic substrates **94**, providing target substrates **95** in 91% ee (Scheme 28c). Formation of the major product could be explained through a (Z)-enolate in a Zimmerman–Traxler transition state. In this model, the B(pin) moiety would have presumably preferred to be in the pseudoequatorial position in a tether formed by the alkyl chain connecting the dienone and the enolate. The structure of the substrate predominantly influenced the diastereomeric ratio of this product. This idea was confirmed by the reaction with the achiral P(OEt)₃ ligand, where the dr was similar to that in the enantioselective case. Similarly, Lin has also performed borylative cyclization.⁸⁹

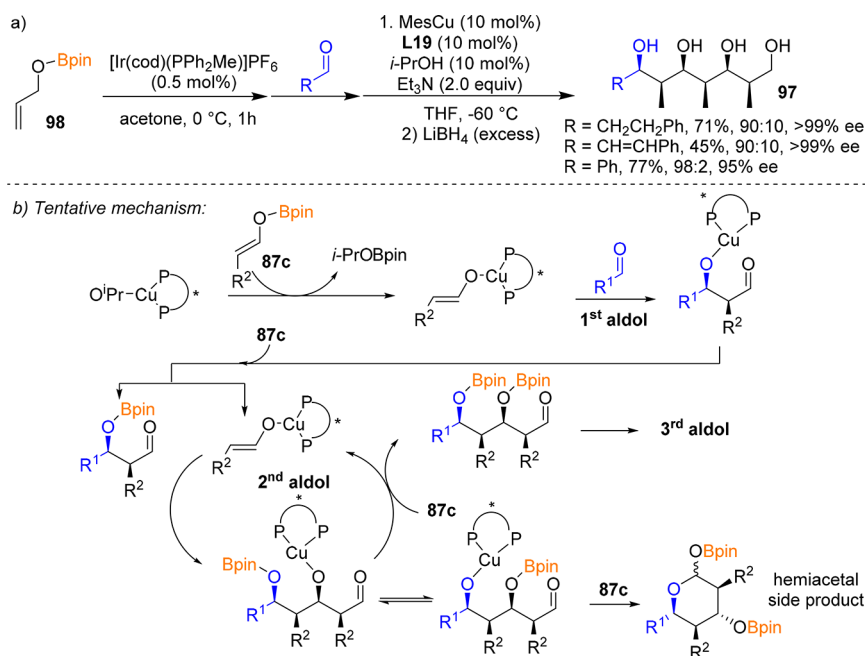
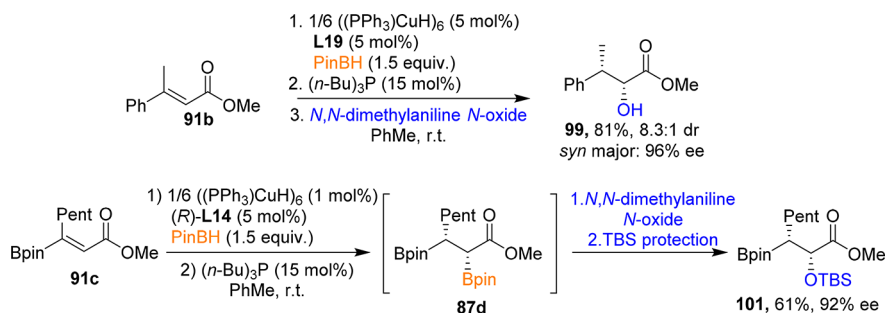
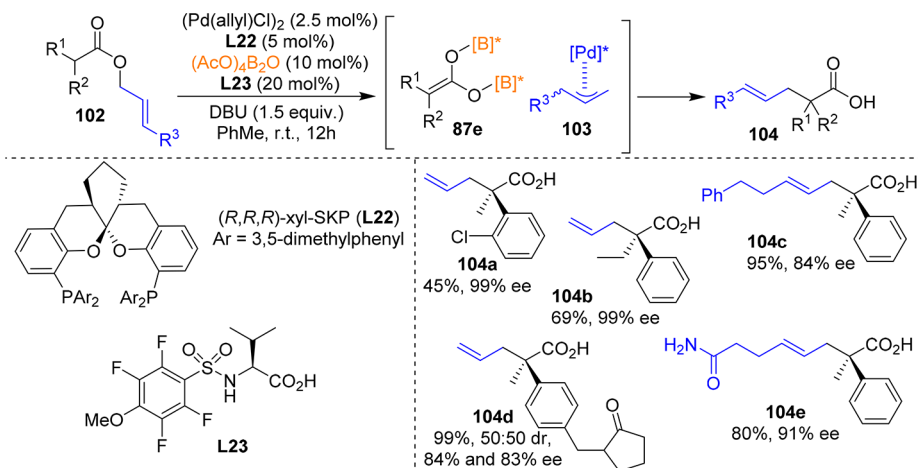
Hayashi reported a Rh-catalyzed CA of organoboron species followed by aldol reaction with an aldehyde. An NMR study revealed that the aldehyde is crucial for the catalytic cycle, as it presumably shifts the catalytic cycle to the formation of the desired product **96** by trapping the Rh enolate to form Rh aldolate, which is consecutively transmetalated to 9-BBN-aldolate by thus reforming the organorhodium reactive species.⁹⁰ The reaction on cyclic acceptors **1a,b** was catalyzed

by [Rh(OMe)(cod)]₂ and (*S*)-BINAP (**L21**). Various electrophiles (MeOH, MeOD, aldehyde, allyl bromide) reacted with the resulting B enolate **87b**. Upon reaction with propanal, the desired products **96** were obtained highly enantiomerically enriched (ee up to 98%, Scheme 29). Notably, the reaction required the use of 9-BBN, as it did not proceed with catecholboranes, pinacolboranes, or cyclotriboroxanes.⁹¹

Scheme 29. Aldol Reaction of Boron Enolate **87b** and Propanal



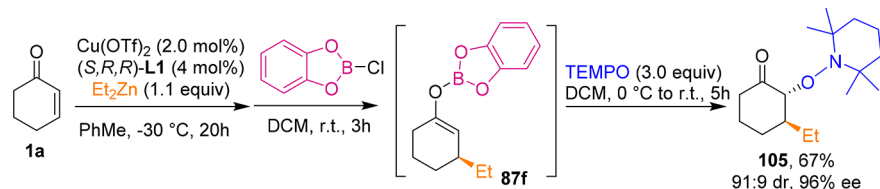
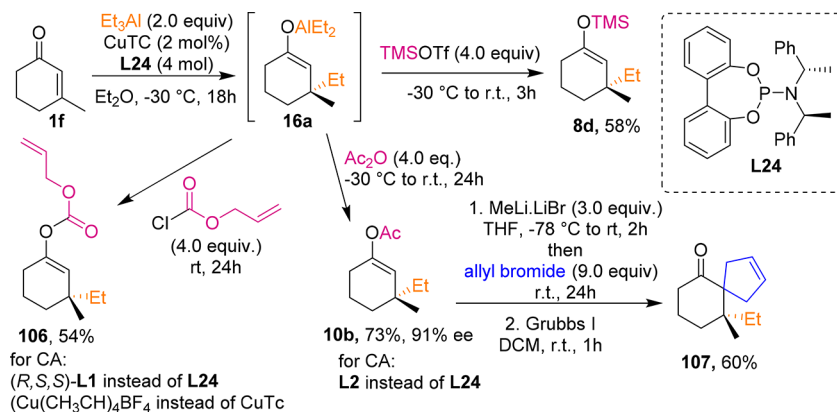
Kanai used cooperative Ir and Cu catalysis to expand this strategy for the enantioselective iterative synthesis of the challenging 1,3-polyol units **97** (Scheme 30). The isomerization of allyloxyborate **98** to boranyl enolate **87c** took place in a separate flask by using preactivated [Ir(cod)(PPh₂Me)]-PF₆. It was followed by an aldol reaction catalyzed by (MesCu)/(*R*)-DTBM-SEGPHOS (**L19**) in the presence of

Scheme 30. (a) Iterative Synthesis of **97** and (b) Proposed Mechanism of the Multi Aldol Reaction and Formation of a Side ProductScheme 31. Asymmetric Reduction of Esters **91b,c** with Pinacolborane, the Transformation of the Boron Enolate to α -Boronyl Ester **87d**, and Oxidation to Alcohols **99** and Protected **101**Scheme 32. Intramolecular Asymmetric Synthesis of α -Allyl Carboxylic Acids **104** from α,α -Disubstituted *O*-Allyl Esters **102**

isopropyl alcohol and starting material. The major difference among the single-, double-, and triple-aldol reactions was the implementation of a B(pin) moiety instead of B(pro) by which the transmetalation step was facilitated. The addition of Et₃N prevented the undesired formation of hemiacetal, and high

yields of the triple-aldol products were obtained after reduction.⁹²

The formation of boron enolates **87** have been thoroughly explored, and it has been established that the thermodynamically driven formation of an O-bound isomer is usually

Scheme 33. Oxidation of B Catecholboronates **87f** to Alkoxyamine **105**Scheme 34. Trapping Reactions of Al Enolate **16a**

predominant, due to the formation of a strong B–O bond. In previous years, however, C-bound boron enolates or α -borylcarbonyl compounds had been shown by spectroscopic methods and later prepared and used in synthesis while mainly tetracoordinate MIDA boryl species were employed, but little to no stereoselectivity was observed.^{93–95}

Chiu and co-workers showed that asymmetric CuH-catalyzed 1,4-reduction of acyclic α,β -unsaturated esters **91b** or amides with pinacolborane followed by subsequent addition of catalytic amount of phosphane affords the corresponding α -boryl esters/amides (Scheme 31). On addition of an anhydrous oxidation step (*N,N*-dimethylaniline *N*-oxide) to the reaction sequence, they were able to prepare the desired α -hydroxy esters **99** in good diastereoselectivities with enantioselectivity up to 96% ee for the major isomer. The reaction is initiated by an asymmetric reduction using a chiral phosphine CuH complex, which generates a chiral O enolate. This enolate can be diastereospecifically isomerized to the corresponding C enolate. Mechanistic studies showed that copper plays a role in the O to C isomerization process. Boron enolates **87** can undergo transmetalation to Cu enolate **100** that equilibrate by O/C isomerization. Phosphine influences the reaction most likely as a ligand to copper by modifying the relative stabilities of O-bound and C-bound Cu enolate **100**. *E/Z* isomerization of the enolate was also observed. The reactivity of prepared C-bound enolates has been shown by a Suzuki–Miyaura cross-coupling reaction as well.⁹⁶

Kanai described the asymmetric synthesis of α -allyl carboxylic acids **104** having an α -quaternary stereogenic center by a Pd- and B-cocatalyzed enantioselective coupling reaction (Scheme 32). Under the reaction conditions α,α -disubstituted O-allyl esters **102** ionized to form the prochiral π -allyl Pd complex **103** that served as an electrophile and carboxylic acid derived B enolate **87e** as a nucleophile. The desired chiral carboxylic acids **104** could be prepared with up to 99% ee in moderate to high yields depending on the character of the substituents attached on α -C. Although the studied reaction

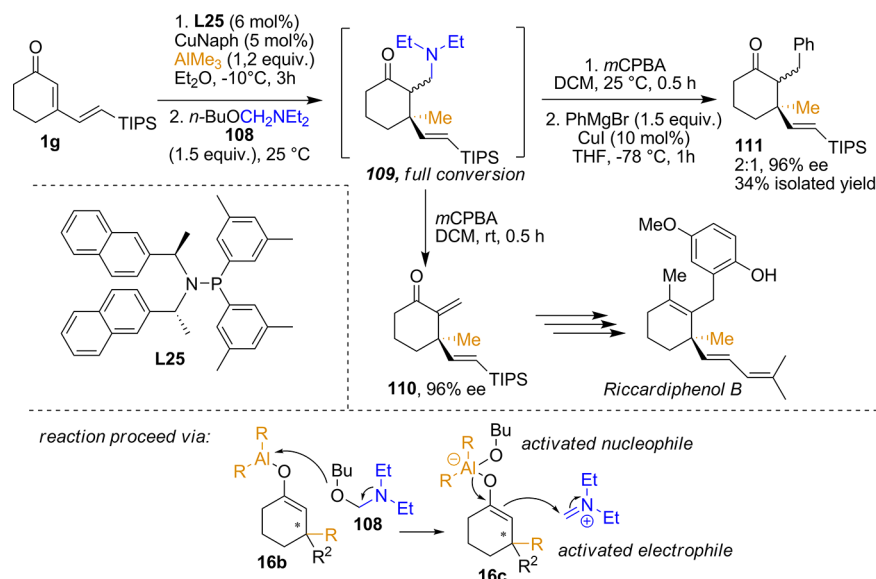
resembles an Ireland–Claisen rearrangement, the intermolecular nature of the allylation step was proved in a test reaction where an equimolar mixture of two starting esters was used. This reaction proceeds intermolecularly between the appropriate carboxylic acid and allyl acetate as well, but the enantioselectivity is lower in comparison to the intramolecular process.⁹⁷

Renaud and Studer have studied the oxidation of B catecholboronates **87f** with TEMPO (Scheme 33). As nitroxides can oxidize the organometallics to the corresponding C-centered radicals that can be irreversibly trapped by the other molecule of oxide to form alkoxyamines, the authors hypothesized that a similar reaction could occur in the case of B-enolates **87**. Indeed, the Cu-catalyzed CA of organozinc followed by transmetalation to B-enolate **87f** and subsequent oxidative trapping by TEMPO afforded alkoxyamine **105** in good yield with high *trans* diastereoselectivity and 96% ee.⁹⁸

Aluminum Enolates. Al enolates **16** represent intermediates that are somewhat less linked with tandem processes due to their high stability, which can be explained by the strong Al–O bond. Their problematic reactivity can be overcome if there is a possibility to coordinate the enolate with an incoming electrophile (aldehyde or imine) or to in situ form alanate by adding a stoichiometric amount of MeLi.

Alexakis and co-workers extensively studied the trapping of Al enolates **16** generated by asymmetric CA of organoaluminum species. Similarly, as with Zn enolates **2**, Al enolates **16** can be O-acylated as well.³⁵ Enolate O-acylation was implemented in the functionalization of highly enantioenriched Al enolates **16a** having quaternary stereogenic centers. The desired trapped products **10** and **8** were obtained in good yields without any decrease in enantioselectivity (Scheme 34).⁹⁹ The O-trapping methodology represents a logical step toward enolate trapping, as it allows re-forming the enolate in a separate step under different reaction conditions. This strategy also demonstrated the value of Alexakis' SimplePhos ligand family in CAs with organoaluminum reagents.¹⁰⁰

Scheme 35. Multistep Synthesis of 111 and 110, Which Is a Precursor to Riccardiphenol B

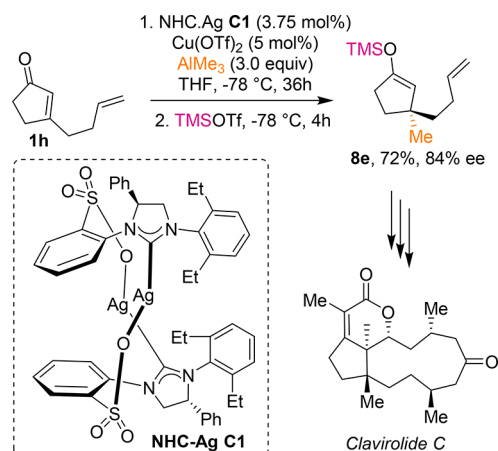


Alexakis studied the direct trapping of in situ generated Al enolates **16** at their α -C position in 2013. On initiation by Cu(I)-catalyzed CA of organoaluminum reagent to cyclic Michael acceptors **1g**, the resulting Al enolates **16b** were successfully trapped by aminal **108** (Scheme 35). The intermediate **109** was subsequently oxidized to Michael acceptor **110** with an exocyclic methylene group, on which another CA was performed. After four consecutive steps in one pot, the desired products **111** were prepared in good yields with an excellent enantiomeric purity of up to 98% ee. A reactive electrophile was generated from aminal **108** by Al enolate **16b** itself by *n*-BuO transfer to Al, thus forming aluminate **16c**.¹⁰¹ This synthetic route was then applied to the catalytic enantioselective total synthesis of sesquiterpene riccardiphenol B.¹⁰² The CA of alkylaluminum species to cyclic Michael acceptors having an exocyclic double bond followed by enolate trapping was studied as well but with inferior results.¹⁰³

Brown and Hoveyda studied silylation of the Al enolate **16** generated by Cu-catalyzed ACA of Me₃Al to β -substituted cyclopentenone **1h**, which was used as one of the steps in the total synthesis of clavirolide C (Scheme 36). The desired silyl enol ether **8e** was prepared in 72% yield and 84% ee by using the chiral NHC-Ag catalyst **C1**.¹⁰⁴

Enolate **16** participated in the asymmetric synthesis of tetracyclic benzo[*d*]xanthene. After the CA of Me₃Al to α -substituted Michael acceptors **1e**, the Al enolate **16d** was used in an alkylation reaction (Scheme 37a).⁹⁹ The low reactivity of the enolate **16d** has been overcome by the addition of MeLi to form the ate complex and by the addition of the highly polar solvent additive HMPA. The reaction went well with benzylic and propargylic iodides and even tolerated functionalized substrates. The desired products **112** and **114** were prepared in good yields with (10–7):1 dr. Further functionalization of the α side chain was shown by a Ru(III)-catalyzed [2 + 2 + 2]-cyclootrimerization reaction.

Al enolates **16d** prepared by ACA of Me₃Al to 2-methylcyclohexenone **1e** were used in a subsequent aldol reaction as the key step in the total synthesis of sesquiterpene (+)-9-oxoeuryopsin (Scheme 37b). Alexakis' conditions were employed for the Cu-catalyzed CA of Me₃Al, and the Al

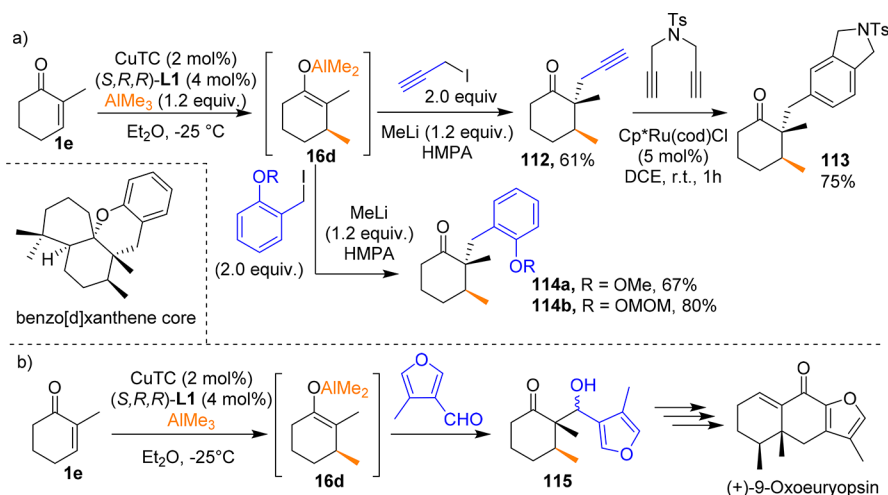
Scheme 36. Silylation of Al Enolate 16 Formed by Cu-Catalyzed ACA of Me₃Al to 1h

enolate **16d** was trapped in an aldol reaction with furan-3-carbaldehyde.¹⁰⁵

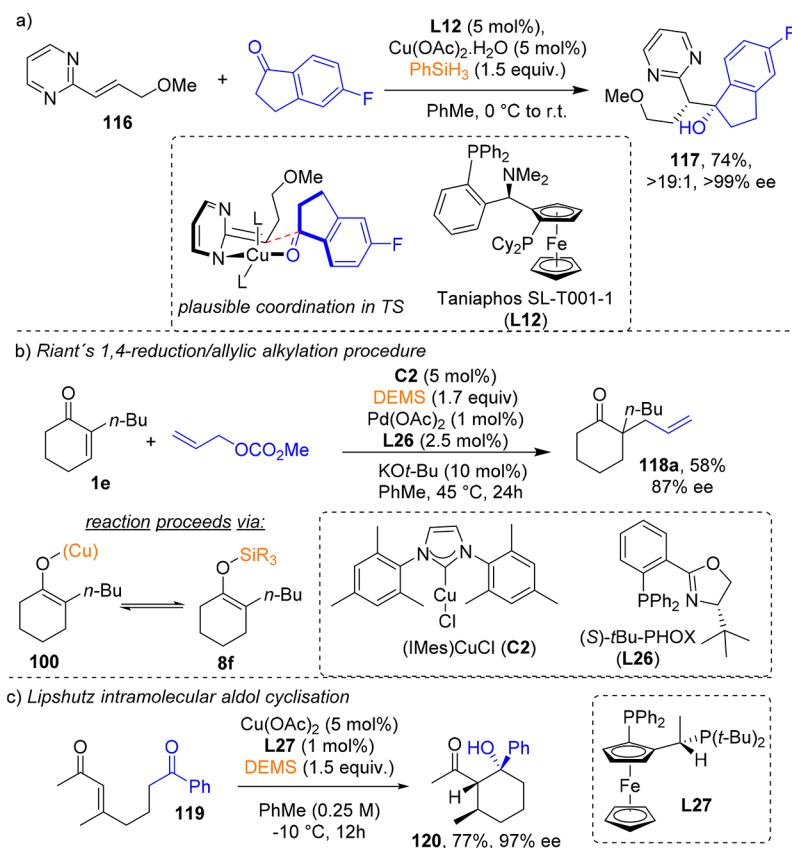
Silyl Enol Ethers. Copper hydride catalyzed conjugate reductions of Michael acceptors generate copper enolates **100**, which most likely transmetalates to silyl enol ethers **8f** if silanes are used as stoichiometric hydride sources.¹⁰⁶ These transient copper (**100**) or silyl enolates (**8f**) can engage in electrophilic trapping reactions. The research teams of Lam,^{107,108} Shibasaki,⁸⁵ Riant,^{109–113} Lipshutz,¹¹⁴ and Chiu independently described tandem processes using this strategy (Scheme 38).^{115,116}

A one-pot sequential Cu-catalyzed asymmetric conjugate reduction, followed by Pd-catalyzed arylation of the silyl enol ether **8**, serves as an example for tandem reactions including a Cu- and Pd-catalyzed step (Scheme 39).¹¹⁷ Diphenylsilyl enol ether **8g** is more reactive than the corresponding TMS derivative **8**. The possible fluorodiphenyl siloxane that could be generated in situ should be even more reactive due to the increased Lewis acidity of the silicon center; hence, high yields of the products could be obtained. Various sources of fluoride anions were tested, from among which CsF afforded the product **121a** in good yields while the formation of desilylated

Scheme 37. (a) Alkylation of Al Enolate **16d** with Propargyl Iodide or Benzylic Iodide and Cyclization to **113** and (b) Aldol Reaction in the Synthesis of a Key Intermediate for (+)-9-Oxoeryopsin



Scheme 38. CuH-Initiated Functionalizations of Enolates

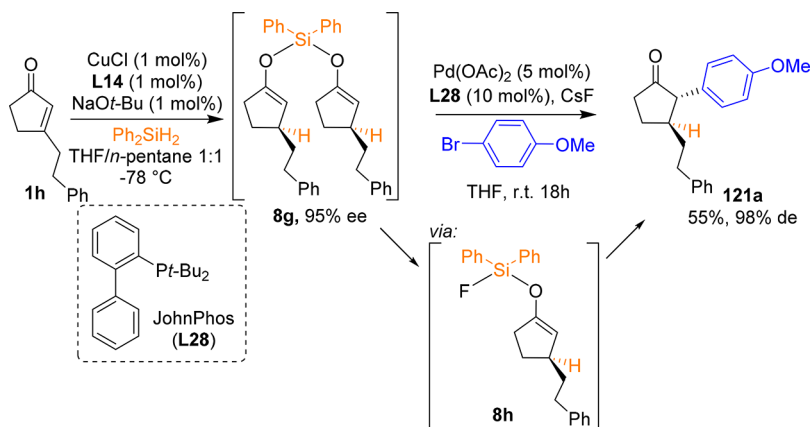


byproducts was minimized. The ligand JohnPhos (**L28**) commonly used for cross-coupling reactions provided the best results here as well. Using THF was crucial for the success of the reaction, as it was much slower in toluene even at elevated temperatures. However, THF negatively affected the enantioselectivity of the reduction (70% ee), as the conjugate reduction required a nonpolar solvent to maintain high enantioselectivity. Therefore, a mixed reaction medium of THF and *n*-pentane was used.

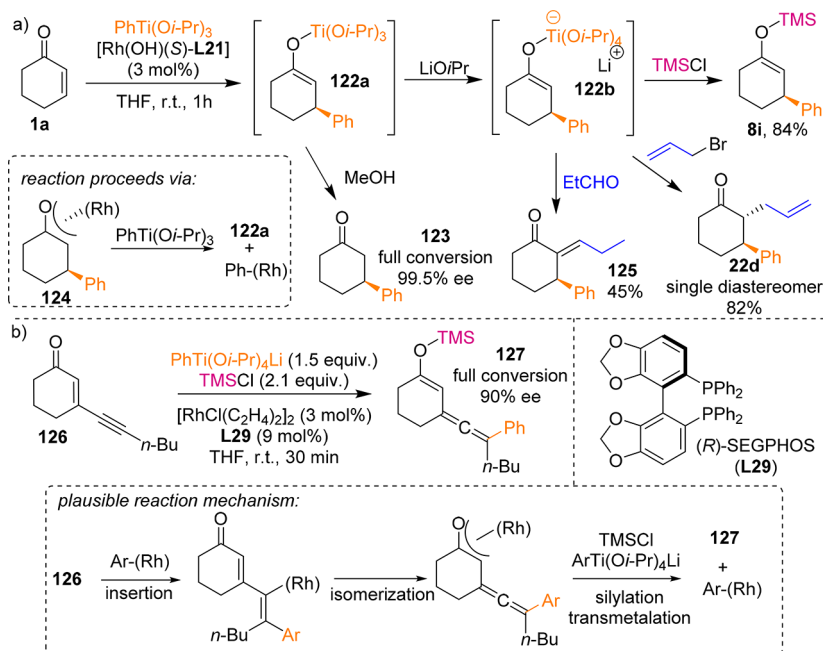
Titanium and Zirconium Enolates. Hayashi realized pioneering studies of Rh-catalyzed CA of organotitanium

species. While having inspiration in Rh-catalyzed 1,4-additions of boronic acids, he established a methodology employing the [Rh(OH)((S)-L21)] catalytic system with aryltitanium reagents. The desired products **123** of CA were obtained with ee values of up to 99.8% for cyclic and acyclic substrates. Titanium enolate **122a** then participated in tandem allylation, TMS protection, or aldol condensation after being further activated by LiO-*i*-Pr to provide **122b** (Scheme 40a). Surprisingly, the CA did not occur in the initial presence of Li salt. NMR studies showed that the catalytic cycle consists of two stages. First, there is a transmetalation of the aryl group

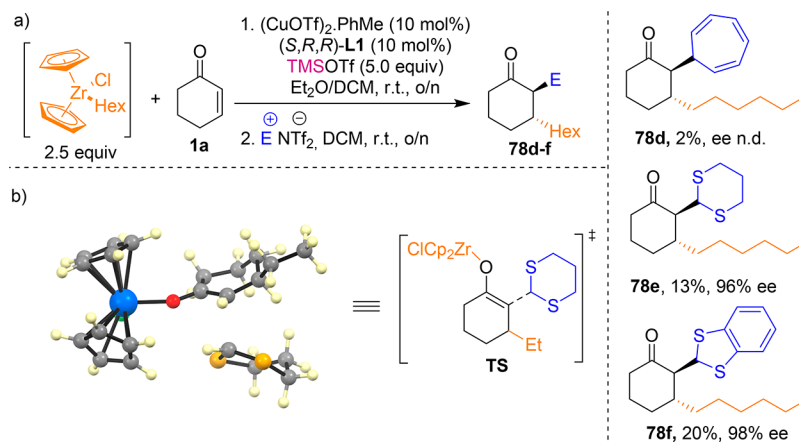
Scheme 39. Tandem Cu-Catalyzed Conjugate Reduction of 1h Followed by Pd-Catalyzed Coupling of the Silyl Enol Ether 8g



Scheme 40. (a) Rh-Catalyzed CA of Phenyltitanium Species to Cyclohexenone 1a, Followed by Trapping Reactions, and (b) Synthesis of Allene 127 by CA and Trapping by TMSCl



Scheme 41. CA Followed by Trapping of the Zr Enolate 128 by Cations



from titanium to the (oxa- π -allyl)rhodium intermediate 124 to form aryl-rhodium species and titanium enolate 122a. Second,

the insertion of enone into aryl-rhodium species occur, thus providing an (oxa- π -allyl)rhodium complex.^{118,119} Hayashi also

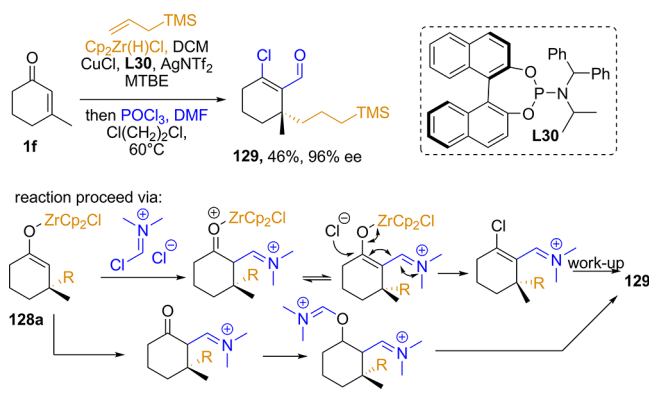
described Rh-catalyzed 1,6-addition of the premade titanate $\text{ArTi}(\text{O-}i\text{-Pr})_4\text{Li}$ species to compound **126** followed by TMS trapping of the resulting enolate, affording axially chiral allenenes **127** in enantiomeric purity up to 93% ee (Scheme 40b). The consecutive use of titanate and TMSCl was crucial for the reaction; otherwise an undesired 1,2-addition prevailed.¹²⁰

Chiral Ti enolates that were prepared by transmetalation of Zn enolates **2** obtained by asymmetric CA were employed in subsequent aldol reactions by Woodward and Welker.¹²¹

Organozirconium reagents have been broadly used in recent years in the field of transition-metal-catalyzed asymmetric CAs and AAs mainly due to the extensive research of Fletcher and co-workers.^{122–124} Our group has reported the first trapping reaction of Zr enolates **128** that was initiated by Cu-catalyzed CA with cationic electrophiles (Scheme 41).¹²⁵ The enantioselective version of this reaction was also studied. Unfortunately, the yields of the trapping products were rather low, due to the low reactivity of intermediate species. DFT calculations also supported the notion of low reactivity of Zr enolates. However, the desired chiral products **78d–f** were prepared in low yields but as sole diastereomers in excellent enantiomeric purities up to 98% ee (Scheme 41a). DFT calculations (B3LYP/6-31G*) revealed the plausible transition state of the enolate trapping reaction (Scheme 41b).

Trapping of Zr enolates by Vilsmeier–Haack reagent was recently reported by Fletcher's group, where both formylation and chlorination occurred.¹²⁶ The unusually structured products **129** were obtained in 41–57% yields with enantioselectivity of up to 98% ee. However, a large excess of the electrophile (up to 20 equiv) was needed for the trapping to occur (Scheme 42). The authors showed the

Scheme 42. Trapping of Zr Enolates 128a with Vilsmeier–Haack Reagent



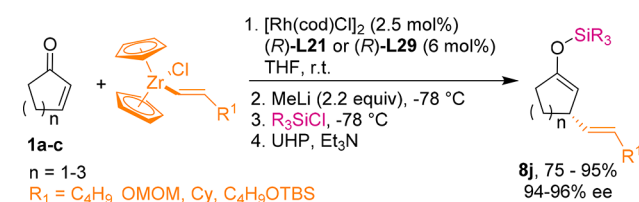
synthetic utility of the prepared products by various Suzuki–Miyaura coupling reactions as well as 1,2-additions or condensations. Two alternative reaction mechanisms were proposed.

Rh(I)-catalyzed CAs of alkenylzirconium species were used by multiple research groups to incorporate the $\text{sp}^2\text{-C}$ in the side chain into various Michael acceptors. To achieve a successful CA, the enolate needs to react with an electrophile as it recycles the Rh catalyst. In all proposed mechanisms the authors considered that after the CA was performed the Zr species shifts the catalytic cycle while re-forming the Rh catalyst by catching the final product in the form of Zr enolate. The reaction thus occurs with Rh enolate instead. Oi and Inoue have shown the asymmetric version of Rh-catalyzed 1,4-

addition of alkenylzirconium species in 2004. They employed an $[\text{Rh}(\text{cod})(\text{MeCN})_2]\text{BF}_4$ or $[\text{Rh}(\text{cod})\text{Cl}]_2/\text{S-L21}$ catalytic system, which provided the desired products in excellent yields and enantiomeric purities of up to 99% ee.¹²⁷

Isolation of the silyl enol ether **8j** was achieved by von Zezschwitz and co-workers, using Oi and Inoue's conditions. To synthesize desired silyl enol ethers **8j** by TMSCl -mediated trapping, the addition of MeLi was required (Scheme 43).

Scheme 43. Trapping of Zr Enolates 128 into Silyl Enol Ethers 8j



Oxidative workup (urea hydrogen peroxide with an excess of Et_3N) provided the corresponding silyl enol ethers.¹²⁸ This strategy was later used in the enantioselective preparation of (R)-sarkomycin.¹²⁹

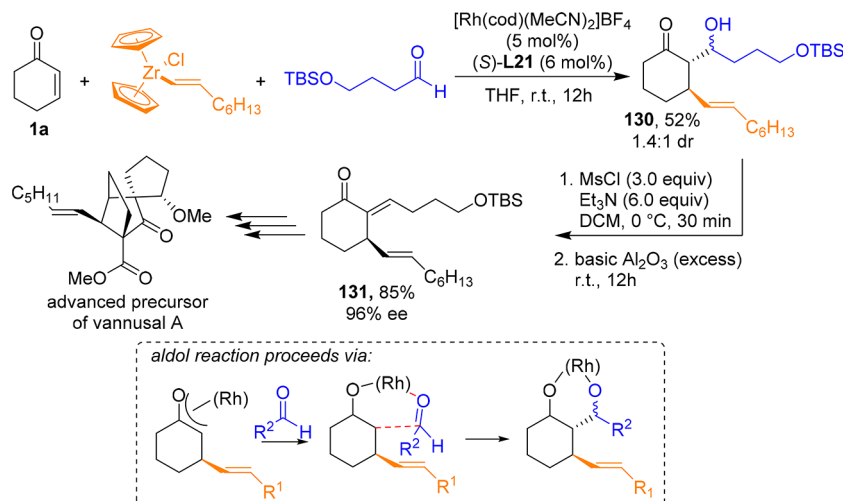
Nicolaou employed Rh-catalyzed asymmetric CAs of alkenylzirconium species with a subsequent aldol reaction in the total synthesis of an advanced intermediate of vannusal A. A one-pot reaction of cyclic and linear Michael acceptor with various aldehydes in the presence of alkenylzirconium by employing the catalytic system $[\text{Rh}(\text{cod})(\text{MeCN})_2]\text{BF}_4/(\text{S})\text{-L21}$ provided the desired products **130** in high yield, but with poor diastereoselectivity. A subsequent elimination reduced the stereocenter at C-2, thus providing product **131** in high ee up to 98% (Scheme 44).¹³⁰ De Clercq has used gaseous formaldehyde as a trapping reagent for the Zr enolate **128**. The resulting products were obtained in 89% yield with a diastereoselectivity of 2.7:1 and an ee of up to 96% by using the catalytic system $[\text{Rh}(\text{cod})\text{Cl}]_2/\text{L21}$.¹³¹

Although indium enolates represent a less explored area in organic chemistry, there was a report of a Rh-catalyzed CA of diarylindium(III) hydroxides to various Michael acceptors.¹³²

BASE-GENERATED ENOLATES

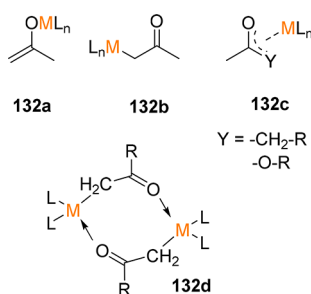
Uncatalyzed reactions of enolates with alkyl halides are common, but the reaction with aryl or alkenyl halide is usually slow and needs a transition-metal catalyst to proceed efficiently. The racemic version of enolate cross-coupling has been known for a while now,¹³³ but the asymmetric version was more difficult to achieve. The complexity of the problem can be demonstrated by the fact that the first asymmetric version of an enolate arylation did not occur until 1998.¹³⁴ To the best of our knowledge, there have been just two reviews on this topic,^{133,135} but no recent and comprehensive coverage of it. Slightly more advanced are allylations of carbonyl compounds via metal-catalyzed transformations.^{136,137} Although this methodology is now approaching maturity, and some excellent results have already been obtained, there are still challenges in the formation of a tertiary stereogenic center next to a carbonyl group. The problem is an even more acidic proton in the arylated product **121**, which can easily racemize under the reaction conditions. To avoid this racemization, less basic conditions or silyl enol ethers **8** can be used to give tertiary stereogenic centers with high

Scheme 44. Tandem CA and Aldol Reaction of Zr Enolate 128: Synthesis of a Key Intermediate of Vannusal A



enantiomeric purities.¹³⁸ For bulky, electron-rich phosphines, only one phosphine is involved in the binding of the metal center of the arylpalladium enolate intermediate.¹³⁹ The structure of the arylpalladium enolate (Scheme 45) also affects

Scheme 45. Structures of the Different Types of Metal–Enolate Bonds

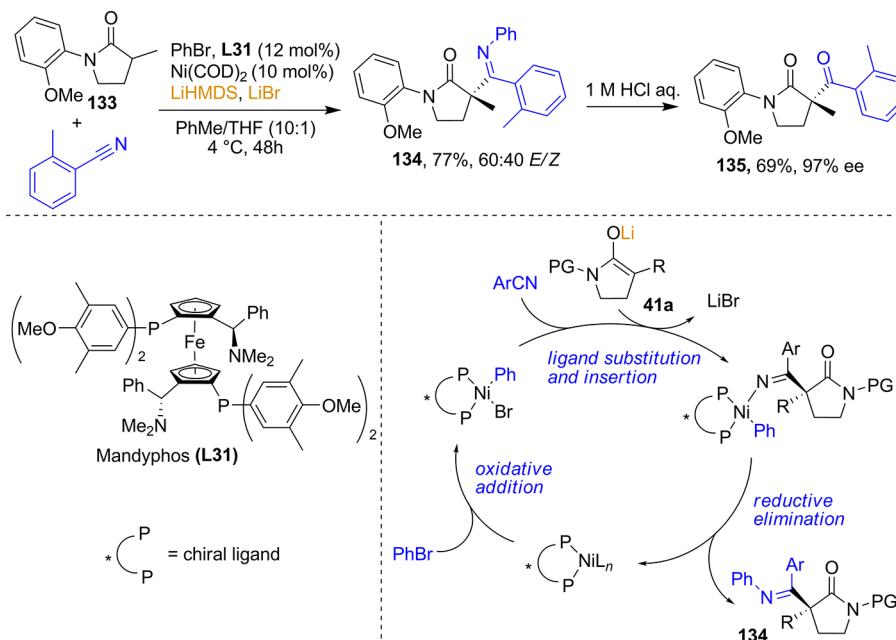


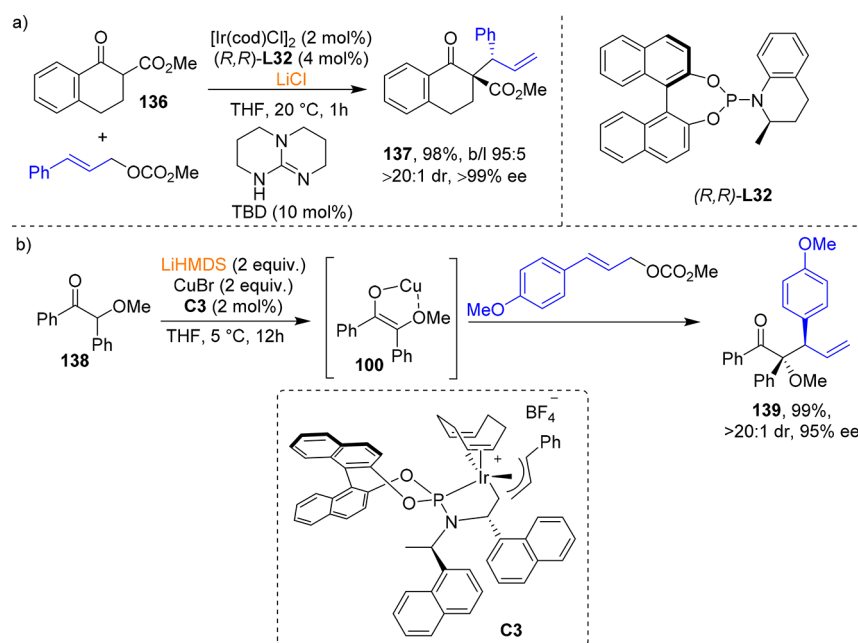
the formation of the product.¹⁴⁰ Transition-metal enolates prefer O-bound (132a) and C-bound (132b) enolates for monocarbonyl compounds,^{141–143} and β -dicarbonyl compounds also prefer η^1 -C- and η^2 -O,O-bound anions (132c).¹⁴⁴ The type 132a is more common for early transition metals; however, it was proposed for palladium in some cases as well.^{145,146} Bridging C–O enolates (132d) have been isolated for Pd complexes.¹⁴⁷

■ ALKALI-METAL ENOLATES

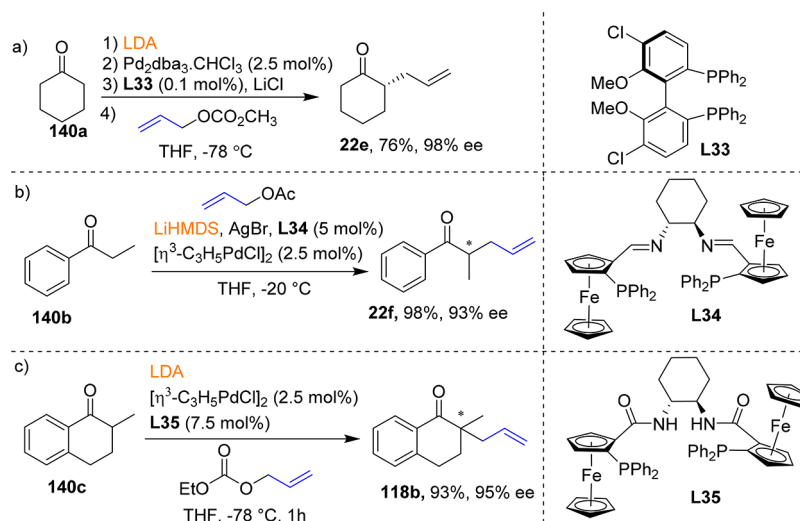
Lithium Enolates. The Stoltz group reported a coupling of lactam 133 derived lithium enolate 41, in the presence of 2-methylbenzonitrile, bromobenzene, and a Ni catalyst (Scheme 46).¹⁴⁸ The Mandypophos ligand (L31) was the most suitable for this reaction. It is possible to obtain the intermediate imine 134 or the corresponding ketone 135 after hydrolysis. The authors demonstrated several synthetic transformations of the lactam products, such as removal of the *o*-methoxyphenyl

Scheme 46. Coupling of Lithium Enolate 41 with Benzonitrile and Bromobenzene



Scheme 47. (a) Enantioselective Ir-catalyzed α -AA of Cyclic β -Keto Ester 136 and (b) Ir/C3-Catalyzed AA of α -Alkoxy Ketones 138

Scheme 48. Allylations of Simple Li Enolates 41



protecting group, reduction with Et_3SiH to the corresponding alcohol, and Baeyer–Villiger oxidation. A plausible mechanism for the enantioselective arylation proceeds via a $\text{Ni}^0/\text{Ni}^{\text{II}}$ redox cycle. The stereodetermining step, in this case, would be the ligand substitution and insertion of the benzonitrile and lactam enolate. Reductive elimination of the Ni^{II} imino complex yields the imine product and regenerates the Ni^0 complex.

Stoltz reported an enantioselective Ir-catalyzed α -allylic alkylation of cyclic β -keto esters 136 (Scheme 47a).¹⁴⁹ The organic base TBD was used for the deprotonation. Lithium salts have a known positive effect on the Ir-catalyzed AAs. In this case, LiBr afforded the product 137 in a 95:5 branched:linear ratio, >20:1 dr, and >99% ee. The catalyst loading could be as low as 1 mol % without any loss of enantioselectivity. Several phosphoramidite ligands provided the products with excellent ee values. Phosphinooxazoline ligands, which are typically used in Ir-catalyzed allylations, did

not give any product under the reaction conditions. The authors also developed a sequential AA followed by Pd-catalyzed diastereoselective decarboxylative AA of TMSE β -keto esters. The oxycarbonyl-protected enolate was revealed upon addition of fluoride and reacted in an allylic alkylation to yield products with quaternary stereocenters.

Highly diastereo- and enantioselective AAs can be achieved for nonstabilized acyclic α -alkoxy ketones 138 (Scheme 47b).¹⁵⁰ The geometry of the enolate 100 is controlled through chelation in the presence of copper. Products 139 containing MOM-protected hydroxy groups were described. The reaction without an additive or with LiCl gave the product in high yields, but unfortunately in very low diastereoselectivity. The reaction has a broad scope with regard to allylic carbonates, as heteroaryl, alkenyl, and alkyl substituents as well as other protecting groups were well tolerated.

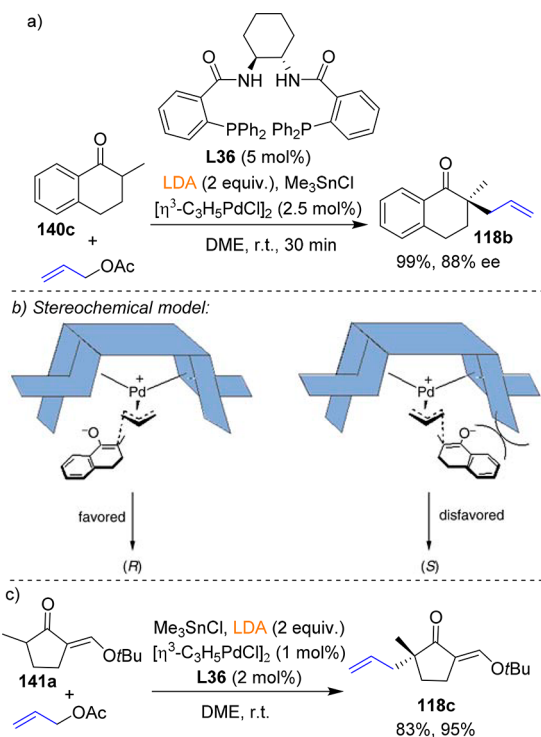
Braun and co-workers described allylations of magnesium **16** and lithium enolates **41** derived from simple cyclic ketone **140a** (Scheme 48a).¹⁵¹ A catalytic system comprising Pd₂dba₃, CHCl₃, and BIPHEP ligand provided the best results. The reaction did not occur without LiCl, as it disaggregates the lithium enolates **41** and enhances reactivity. Unfortunately, the reaction conditions used for cyclopentanone and α -tetralone **140c** gave only modest enantioselectivities.

Inspired by Trost's chiral "pocket" ligands, the group of Hou introduced the ferrocene-based phosphine ligand **L34** (Scheme 48b).¹⁵² LiHMDS as the base and AgBr as a Lewis acidic additive provided the optimum results (98% yield and 93% ee). When α -alkoxy-substituted ketones were used, high enantioselectivities were observed, due to the formation of the Z-chelated enolate **41**.

The ferrocene type ligand **L35** was also studied by the group of Zhu.¹⁵³ Since the ligand **L35** contains two chiral centers and also possesses axial chirality on the ferrocene, it is important to find a matched pair of chiralities. Under the optimized reaction conditions product **118b** was obtained in 93% yield and 95% ee (Scheme 48c). Interestingly, the dihydrate of the ligand **L35** gave a higher ee value in comparison to that for the water-free species, even though lithium reagents react rapidly with water.

The Trost group pioneered asymmetric AA on acyclic nonstabilized ketones in 1999.¹⁵⁴ Trimethyltin chloride was used as a Lewis acid to soften the lithium enolate **41** by transmetalation to the tin enolate (Scheme 49a).¹⁵⁵ 2-Methyl-1-tetralone (**140c**) was used as the substrate, and the reaction was carried out in DME, with LDA as the base. The choice of a lithium base was crucial, since the harder sodium and potassium enolates might react with the palladium catalyst,

Scheme 49. (a) AAs of Sn Enolates Developed by Trost, (b) Stereochemical Model Reprinted with Permission from Ref 155,^a and (c) Synthesis of Hamigeran B Intermediate

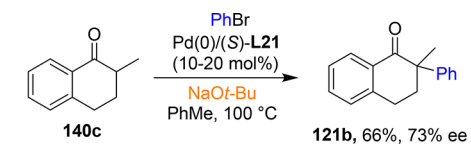


^aCopyright 2005 John Wiley and Sons.

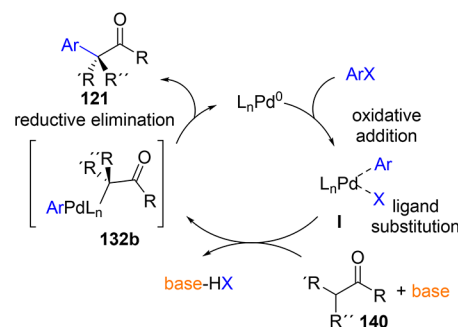
thus promoting undesired side reactions. Some aluminum (**16**), indium, titanium (**122**), and cerium enolates were also tested but provided inferior results. The origin of the absolute configuration of the product was rationalized by using a model of the chiral pocket (Scheme 49b). The geometry of the ligand places the phenyl groups on the phosphorus in space in a manner which favors the approach of the allyl from the Si face of the enolate, resulting in the formation of the R product **118b**. The Trost group also used the asymmetric AA in the synthesis of a key intermediate for hamigeran B (Scheme 49c).¹⁵⁶ The use of *t*-BuOH was essential to make a lithium alkoxide in situ, which presumably affects the nature of the enolate clusters. Hamigeran B was prepared in a 15-step synthesis in 10% overall yield.

Sodium Enolates. Buchwald reported the first asymmetric coupling of enolates to give quaternary stereogenic centers in 1998 (Scheme 50).¹³⁴ The Pd/(S)-**L21** catalytic system

Scheme 50. First Asymmetric Cross-Coupling of 2-Methyl-3,4-dihydronaphthalen-1(2H)-one **140c** in the α -Position and Plausible Catalytic Cycle



Mechanism of arylation of enolates:

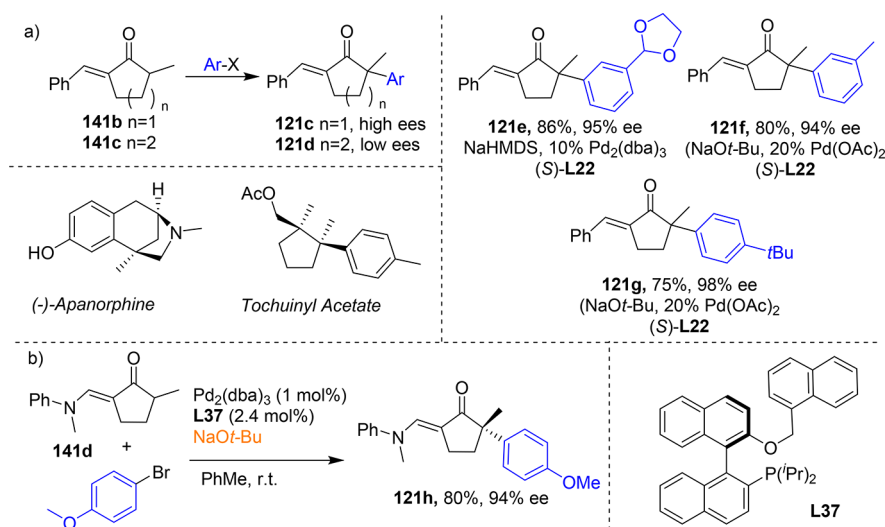


furnished products **121** in modest yields and enantioselectivities in the initial study. NaO-*t*-Bu was used as a base to generate the prochiral enolate, and the reaction required an elevated temperature. A possible catalytic cycle for the palladium-catalyzed arylation of enolates with aryl halides features in the first step a Pd⁰/L complex, which undergoes oxidative addition of the aryl halide, forming the arylpalladium(II) halide complex **I**. The enolate nucleophile then replaces the halide anion in the palladium coordination sphere. Reductive elimination from the palladium enolate complex **132b** then forms the arylated product **121**. Pd⁰ is regenerated and starts a new catalytic cycle.

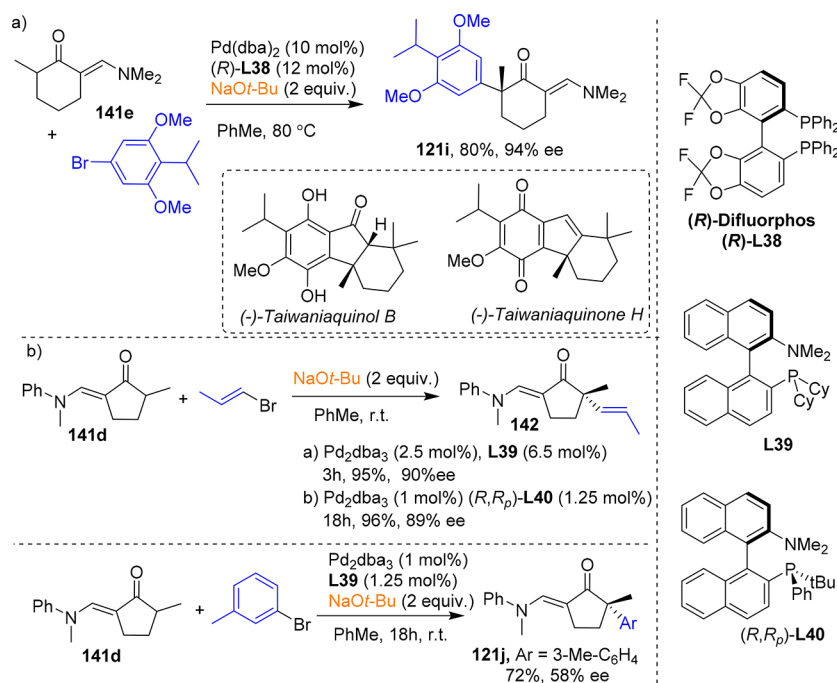
α' blocking groups were used to prevent arylations to both sides. When α' -blocked α -methylcycloalkanes were used, the six-membered-ring species **141c** gave products in low enantiomeric purities (Scheme 51a). Nonetheless, the five-membered-ring species **141b** reacted with aryl halides under Pd/(S)-**L21** catalysis with high enantioselectivity to give products **121** in excellent yields.

Coupling products, optically active 2-methyl-2-arylcyclopentanones, were used as intermediates in the synthesis of natural products (–)-apanorphine and tochuinyl acetate.¹⁵⁷ Later, Buchwald reported the more efficient biphenyl-based ligand

Scheme 51. (a) Cross-Coupling of α' -Blocked α -Methylcycloalkanes **141b,c** and (b) Buchwald's Asymmetric Arylation of **141d** with Pd/L37 Catalyst System



Scheme 52. (a) Enantioselective α -Arylation in the Synthesis toward Taiwaniaquinols H and B and (b) Vinylation and α -Arylation of α' -Blocked α -Methylcycloalkanes **141d**



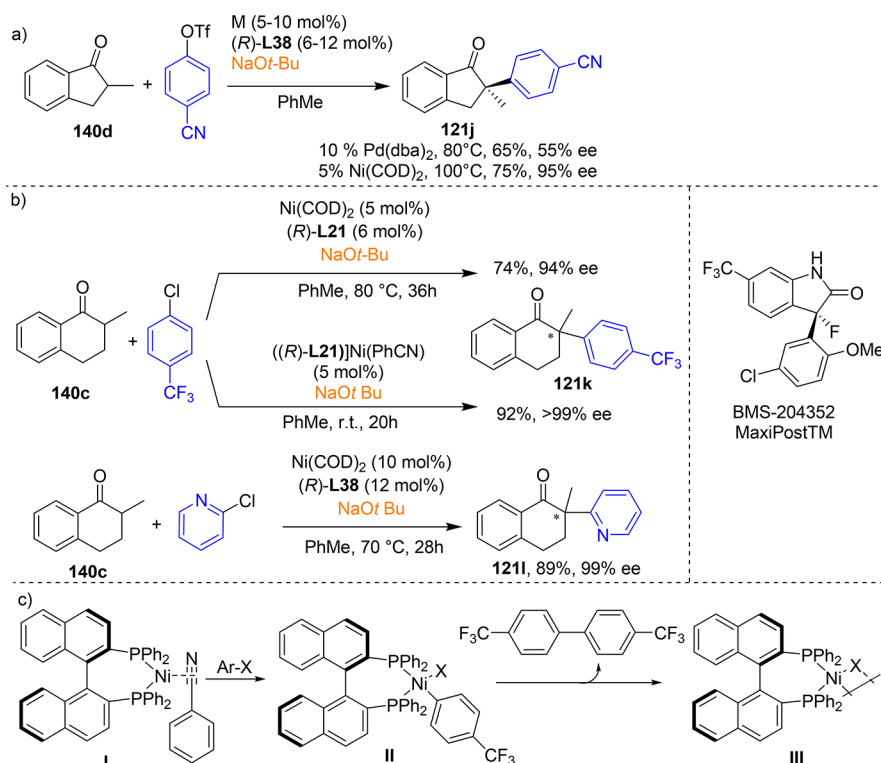
L37 in an asymmetric ketone arylation (Scheme 51b). L37 required only 1% Pd, and the reaction ran efficiently even at room temperature in comparison to 100 °C when (S)-L21 was used. To prevent arylation of the less hindered side of the ketone, they installed a blocking *N*-methylanilinomethylene moiety. The blocking group could be removed in high yield, without any loss of enantiomeric purity.

Hartwig used an enantioselective α -arylation in the synthesis of taiwaniaquinone H and taiwaniaquinol B, which belong to the family of unusual diterpenoids with antitumor activities.¹⁵⁸ Attempts to arylate 2,2,6-trimethylcyclohexanone with an aryl bromide using $\text{Pd}(\text{dba})_2$ and (R)-SEGPHOS (L29) or (R)-difluorophos (L38) were unsuccessful. Using the ketone **141e** with one of the α positions blocked yielded the arylated product **121h** in 80% and 94% ee (Scheme 52a).

Axially chiral ligands were tested in an asymmetric vinylation of ketones (Scheme 52b).¹⁵⁹ With L39 and $\text{Pd}_2(\text{dba})_3$ the product **142** was obtained in 95% yield and 90% ee. The enantioselectivity of the reaction was also strongly influenced by the structure of the vinyl bromide. When an (*E*)-alkene was used, >90% ee was obtained, but with (*Z*)-alkene the ee dropped to 76%. Later, chirality on phosphorus was introduced in the ligand L40.¹⁶⁰ (*R,R*)-L40 and L39 afforded the product of vinylation in high yields and ee values. Interestingly, the diastereomeric ligand (*R,S*)-L40 showed dramatically lower enantioinduction (34% ee). The arylation went much worse, with the best result being 72% yield and 58% ee using ligand L40.

The Hartwig group reported a cross-coupling of enolates derived from methyl indanones **140d** with aryl triflates

Scheme 53. (a) Cross-Coupling of Enolates Derived from Methyl Indanones **140d** with Aryl Triflates, (b) Arylation of Tetralone **140c** with Chloroarenes, and (c) Results of Mechanistic Studies Performed by Hartwig



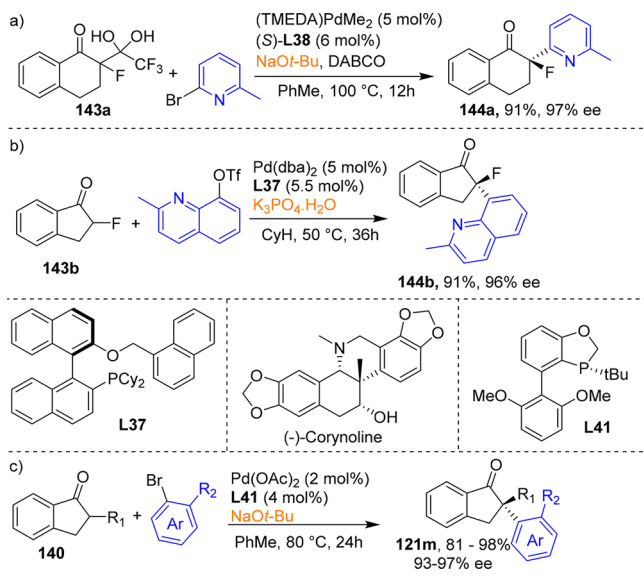
(Scheme 53a).¹⁶¹ The authors found out that a small dihedral angle on the backbone of the chiral bisphosphane was important for the success of the reaction, with (R)-**L29** and (R)-difluorophos (**L38**) giving the highest enantioselectivities. Interestingly, a Pd catalyst was efficient for electron-rich aryl triflates, and a Ni catalyst was necessary for electron-poor substrates. Aryl bromides reacted under Pd catalysis in reasonable yield and enantioselectivity, but iodotoluene reacted poorly, which is due to an inefficient transmetalation and reductive elimination sequence with Pd(**L29**)(Tol)(I). The reactions of α -methylindanone **140d** with aryl triflates were faster than those of α -methyltetralone **140c**. The products were obtained in high yields and up to 95% ee. The arylation of α -methylindanone and α -methyltetralone **140c** with electron-poor triflates and the **L38**/Pd catalytic system afforded the product with only 55% ee, but the same reaction except with Ni(COD)₂ gave the product in 95% ee.

Arylation of tetralones **140c** and indanones **140d** with chloroarenes catalyzed by Ni was later given more attention (Scheme 53b).¹⁶² Under the given conditions, aryl chlorides were more reactive than the aryl bromides, iodides, and triflates. The bidentate ligand (R)-**L21** gave the best results for aryl chlorides and (R)-**L38** for heteroaryl chlorides. If the pregenerated complex [(R)-**L21**]Ni(η^2 -NC-Ph) was used, the products **143** were obtained at room temperature in high yields and enantiopurities of greater than 99% in the majority of the cases. This complex adds chloro- and bromoarenes more rapidly than [(R)-**L21**]Ni(COD), which dissociates slowly and slows the oxidative addition of haloarenes to Ni(0) species. Several mechanistic studies using (R)-BINAP (R)-**L21** were conducted to identify the source of the high enantioselectivity, as well as to explain the different reactivity of chlorides and bromides. Two Ni species were detected by ³¹P NMR—[(R)-

L21]₂Ni and [(R)-**L21**]Ni(COD). [(R)-**L21**]₂Ni proved not to be the active catalyst in the reaction. [(R)-**L21**]Ni(COD) reacts with bromo- or chloroarenes to give products that are thermally labile toward the generation of Ni(I) species in the absence of enolate. It was shown that the catalyst decomposes differently in the two cases. While the chloroarenes gave the product with a consistent ee over time, when bromoarene was used, the catalyst decomposed to a second species that catalyzed the reaction with lower enantioselectivity. Further experiments showed that benzonitrile triggers the oxidative addition of chloroarenes to [(R)-**L21**]Ni(COD). Indeed, this complex **I** was isolated and upon addition of chlorobenzotrifluoride gave complex **II** in 5 min (Scheme 53c). When stoichiometric reactions of **II** with the sodium enolate of 2-methyl indanone **140d** reacted, similar results were obtained for the stoichiometric, and catalytic reaction of aryl chlorides. On the other hand, for aryl bromides the values for the stoichiometric reaction were much higher. This finding supports the previous statement that **II** decomposes into a less selective and reactive catalyst for aryl bromides. Complexes **II** decomposed into complexes **III**, which were used in the catalytic reactions. The complexes did not catalyze the reactions at room temperature and at 80 °C gave poor results. Thus, the loss of reactivity and selectivity can be assigned to the accumulation of the Ni(I) species. Using this methodology, Starrett synthesized the α -aryl- α -fluoro carbonyl derivative BMS-204352.¹⁶³

Hartwig's group reported an enantioselective Pd-catalyzed coupling of α -fluoroketones **143** with bromoarenes (Scheme 54a).¹⁶⁴ In some cases, the addition of potassium trifluoroacetate to 2-fluoroindanones **143** gave higher yields of products. The authors speculated that higher yields could be attributed to the promoted formation of a Pd enolate complex.

Scheme 54. (a) Coupling of α -Fluoroketones **143a** with Bromoarenes, (b) Coupling of α -Fluoroindanones **143b** with Bromoarenes, and (c) α -Arylation of **140** by Ortho-Substituted Aryl Halides

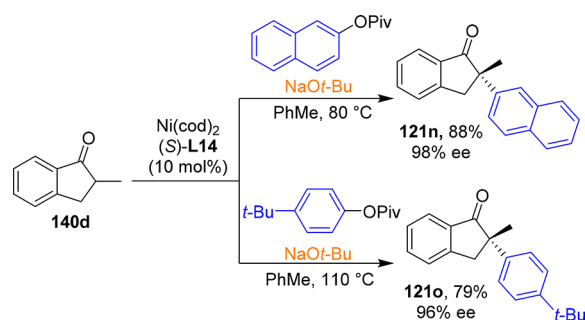


This finding allowed for the coupling of electron-poor arenes with Segphos **L29** in high yields as a complementary method. α -Fluoroenolates have a different reactivity in comparison to their nonfluorinated analogues. α -Fluoroenolates are less nucleophilic and more challenging to generate because they are unstable under basic conditions and prone to protode-fluorination. Furthermore, reductive elimination of fluoroalkyl Pd complexes is usually slower than that of nonfluorinated analogues.^{165–167} Despite these concerns, aryl bromides and triflates gave the corresponding products **144** in high yields and enantiomeric purities when difluorophos **L38** was used as a ligand. The identity of the Pd source and the addition of 5 equiv of DABCO to the NaO-*t*-Bu were key to effective coupling conditions. 2-Fluoroindanones **143b** can also react with aryl bromides or aryl triflates using the binaphthyl-based ligand **L41** (Scheme 54b).

The Tang group developed α -arylation of sterically hindered substrates, which is still a challenging task to tackle.¹⁶⁸ Bulky and *ortho*-substituted aryl halides tend to give products in low yields and ee values. A sterically hindered BI-DIME type ligand performed well in this reaction and gave products in 73–93% yields and ee values in the range of 53–95% (Scheme 54c). Chiral monophos ligands **L41** afforded the best results (93% yield, 95% ee). The synthetic utility was demonstrated in the synthesis of (–)-corynoline and related compounds.

An enantioselective cross-coupling using aryl esters was also reported (Scheme 55).¹⁶⁹ C–O electrophiles have been gaining more attention in recent years due to the popularity of aryl sulfonates and the low toxicity of phenols in comparison to aryl halides. Aryl sulfonates are very reactive because of easy C–O oxidative addition. In contrast, the aryl esters require a higher temperature for the cleavage of the unreactive C–O bond. There are more possibilities of where the bond can be broken in the reaction and the apparent problem of hydrolysis under basic conditions. The Ni(cod)₂ and (S)-**L14** catalytic system performed the best, providing the arylated product **121** in an 88% yield and 98% ee; other bidentate diphosphine ligands such as (S)-**L21** and (S)-**L38** also gave good results. It

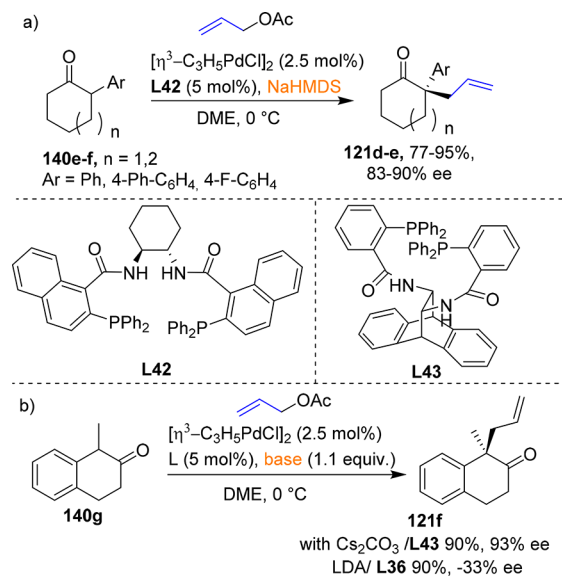
Scheme 55. α -Arylation of 2-Methylindanone **140d** by Aryl Pivalates



is known that the use of extended π systems is advantageous in catalytic C–O bond cleavage processes.¹⁷⁰ This is probably due to the ease in which η^2 coordination to the metal center and the Meisenheimer-type complexes makes extended π systems more reactive than regular arenes.

Trost developed a selective α -allylation of α' -unblocked enolates derived from aryl ketones **140e,f**.¹⁷¹ With a sterically demanding ligand, the use of a sodium enolate was the best fit for approaching the chiral pocket and gave the products in yields up to 95% and enantiomeric purities of up to 92% ee (Scheme 56a). Various cycloalkanones **140** provided products

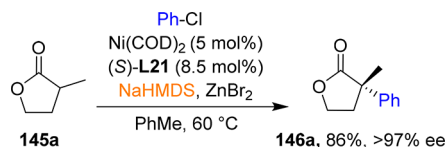
Scheme 56. Allylation of Unblocked Ketones **140e–g** Using Trost Ligands



with good ee values. Diverse functional groups and sterically more demanding aryl substituents were also tolerated. Unfortunately, the reaction conditions could not be applied to tetralones **140g**, for which different conditions were developed (Scheme 56b). The ee values obtained using the standard ligand were dramatically affected by the base used. With LDA and cesium carbonate the opposite enantiomers of the product **121** were obtained, thus indicating the importance of the structure of the ion pair, which reacts, not just the enolate itself.

The arylation of lactones **145** was studied by Buchwald's group as well (Scheme 57).¹⁷² The Ni(0)/(S)-**L21** catalytic system was used for α -substituted γ -butyrolactones **145** with

Scheme 57. Coupling of Lactone 145a with Chloroarenes



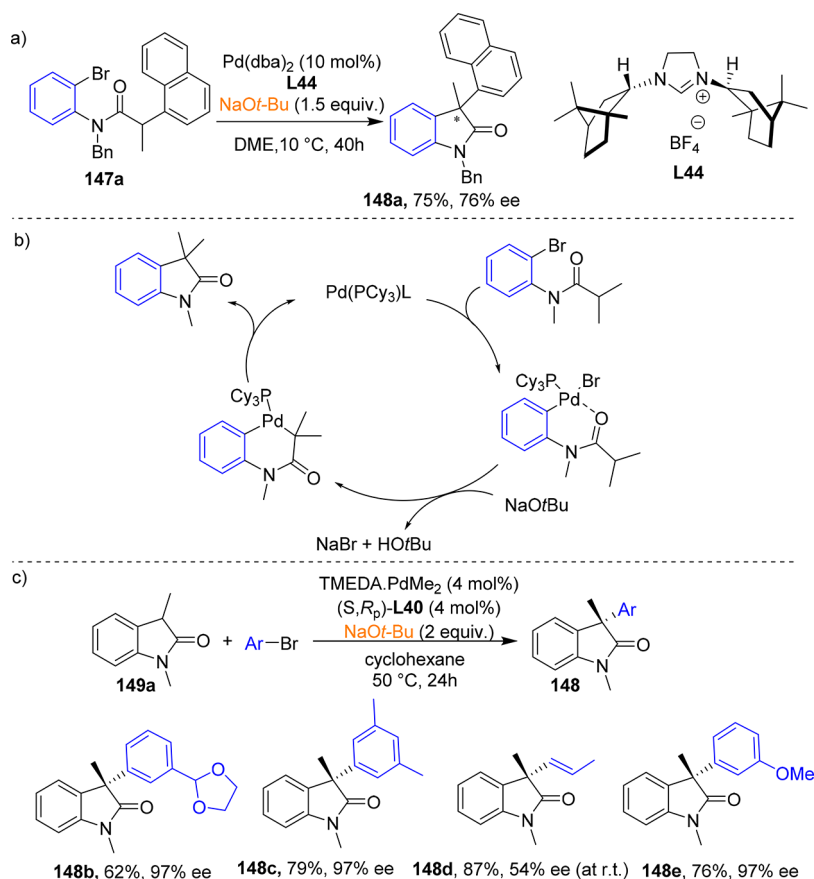
aryl chlorides or bromides. Pd(0)-catalyzed arylation showed worse results regarding yield and enantioselectivity. The reaction rate and yield of **146a** have further increased with addition of ZnBr_2 . Yields were modest to excellent with a variety of aryl halides. The authors explained this acceleration by proposing that ZnBr_2 acts as a Lewis acid which accelerates bromide abstraction from $(\text{L21})\text{Ni}(\text{Ar})(\text{Br})$ to give a cationic $[(\text{L21})\text{Ni}(\text{Ar})]^+$ species that then undergoes transmetalation more rapidly.

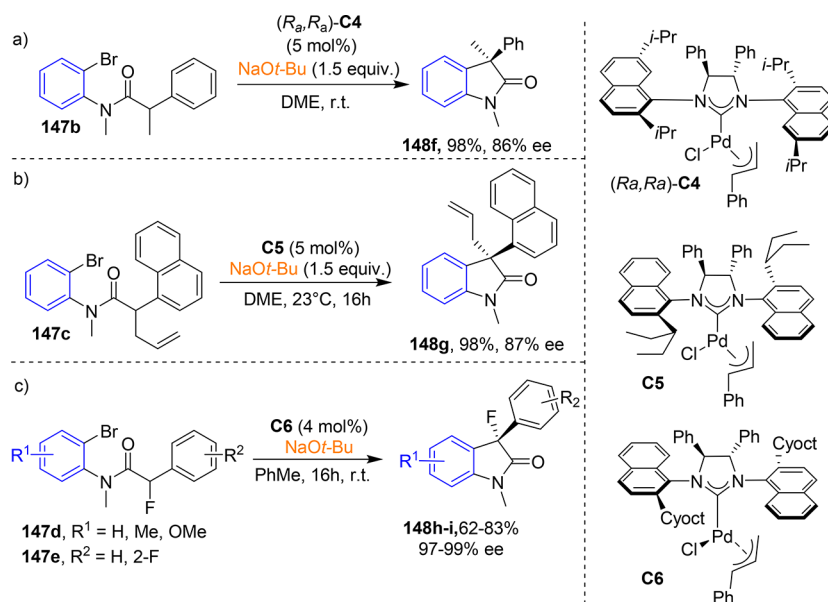
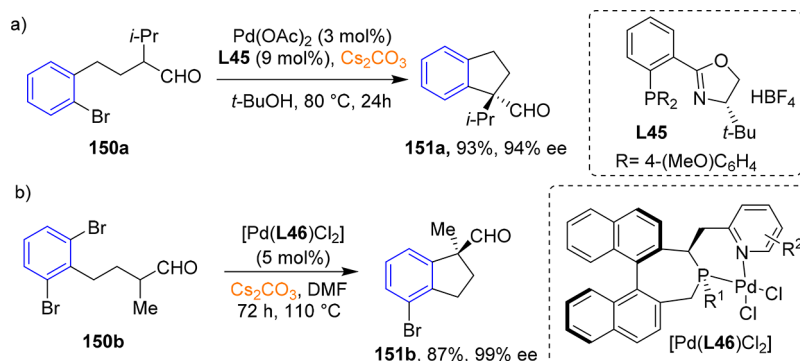
Modest enantioselectivity was obtained in the synthesis of oxindole **148a** from 2-haloanilide **147a**, with a Pd/carbene ligand **L44** system (Scheme 58a).¹⁷³ A careful selection of base, counterion, and solvent was important for the enantioselectivity. Sodium *tert*-butoxide in DME gave the highest selectivity. The use of carbene ligand **L44** enabled product formation at lower reaction temperatures. The reaction of *N*-methyl bromide and *N*-methyl chloride occurred with similar enantioselectivity, thus suggesting that the halide is not present on the complex during the enantiodiscriminating step. However, the reaction with the iodo derivative gave the product with a very small or no enantiomeric excess. The

mechanism of the cyclization is assumed to follow this pattern—oxidative addition of the aryl halide, formation of an arylpalladium amide enolate, and reductive elimination give the product (Scheme 58b).

The ligand mentioned earlier turned out to be very useful in an enantioselective arylation also on 1,3-dimethyloxindoles **149a**, even though it performed poorly in the previous case.¹⁷⁴ All three of the axially chiral ligands gave products **148** with good enantioselectivities and ee values, though that with a chiral stereogenic center on phosphine, (S,*R*_p)-**L40**, gave the best results (76% yield, 97% ee) with TMEDA-PdMe₂ as the Pd source (Scheme 58c). Only biaryl monophosphanes were useful in this transformation. Various α -arylations and α -vinylations were carried out with high yields and enantiomeric purities.

Dorta reported an enantioselective arylation of α -haloanilides **147** using chiral monodentate carbene ligands.¹⁷⁵ Three diastereomeric ligands were tested in a series of reactions, most of which yielded the products in excellent yields. The complex (R_s,R_a)-**C4** gave the highest enantiomeric purities of 80–88% ee for all substrates (Scheme 59a). Later another carbene ligand complex, **C5**, was prepared for enantioselective intramolecular arylation of amides **147**.¹⁷⁶ A bulky 3-pentyl moiety was incorporated on the 2-position of the naphthyl side chain for the reaction (Scheme 59b). The authors envisioned that the new ligand complex **C6** with a cyclooctyl group at the 2-position of the naphthyl moieties could be useful for enantioselective α -arylation leading to enantioenriched 3-fluoro-3-aryl oxindoles **148** (Scheme 59c).¹⁷⁷ This method-

Scheme 58. (a) Intramolecular Synthesis of Oxindole 148a from 2-Haloanilide 147a with the Pd/Carbene Ligand L44, (b) Mechanism of Cyclization of Haloamides 148, and (c) Synthesis of Oxindole 148 by α -Arylation of 149a

Scheme 59. Intramolecular α -Arylation of Amides Catalyzed by Pd Carbene Ligand Complexes C4–C6, To Give Oxindoles 148Scheme 60. Asymmetric Intramolecular α -Arylation of Aldehyde 150

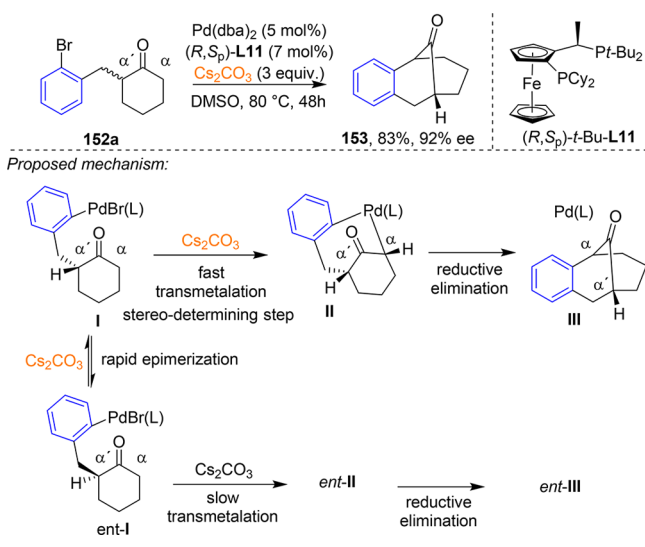
ology would be a complementary method to the catalytic enantioselective fluorination method.¹⁷⁸ The reaction using 5 mol % of the catalyst in toluene proceeded smoothly at room temperature to give the product in 83% yield and 97% ee. The reaction showed an excellent substrate scope—electron-donating and -withdrawing groups on *ortho*-, *meta*-, or *para*-positions led to excellent enantioselectivities and yields at room temperature or at 50 °C. A DFT study was performed to explain the origin of the stereochemistry. A key intermediate—the expected compound before reductive elimination—was optimized. Structural analysis suggests that the intermediate leading to the experimentally favored *S* product **148** is sterically more accessible. The phenyl substituent is in this case under the naphthyl ring. On the other hand, in the intermediate leading to the *R* product **148**, the phenyl ring is under the cyclooctyl ring. The energy difference between them is 4.4 kcal mol^{−1} in favor of the first.

Cesium and Potassium Enolates. Asymmetric α -arylation of aldehydes **150** is a challenging task due to the competitive aldol reaction.¹⁷⁹ So far, there is only one example of the intramolecular α -arylation of aldehydes, which uses phosphaneoxazoline ligand **L45** (Scheme 60a). Interestingly, ligands with bulkier phosphorus substituents gave lower enantioselectivities. A solvent screening revealed *t*-BuOH to

be optimal, instead of DME. Ligand **L45** gave the product in 93% yield and 94% ee. Substrates **150** with α -aryl or α -alkyl substituents were well tolerated, but substrates leading to six-membered rings reacted poorly. Similarly, Mazet developed highly enantioselective intramolecular Cs enolate arylation using binaphthyl ligand **L46** (Scheme 60b).¹⁸⁰

Zhou used a catalytic arylation of ketones **152** to prepare bridged bicycles **153** via dynamic kinetic resolution (Scheme 61).¹⁸¹ A moderately strong base, Cs_2CO_3 , was used to promote rapid and reversible racemization at the α' position of the ketone **152**. Common ligands such as (*R*)-**L21**, (*R*)-**L38**, (*R*)-**L29**, and (*R*)-MeO-biphep led to less than 30% ee of the product, but Josiphos-type ligands **L11** with bulky, strongly donating phosphine groups afforded the product in high ee values. Polar aprotic solvents were crucial for achieving high yields and enantioselectivities due to solvation of the alkaline metal ions of the bases. Cyclopentanone derivatives could be cyclized with excellent ee values, but cycloheptanone and cyclooctanone derivatives reacted poorly under the reaction conditions. The reaction mechanism likely involves an oxidative addition of the racemic starting material to the Josiphos-**L11**-Pd complex, resulting in two diastereomeric complexes, which are in equilibrium under basic conditions. Deprotonation of **I** and transmetalation from the cesium

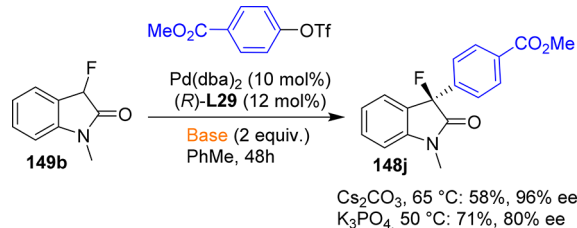
Scheme 61. Intramolecular Arylation of Ketone 152a with Proposed Mechanism



enolate to arylpalladium bromide is much faster than that of ent-I. Finally, reductive elimination takes place from Josiphos- (R,S_p) -*t*-Bu-L11/Pd complexes to form a new C–C bond.

The Shi group introduced a Pd-catalyzed arylative desymmetrization of 1,3-diketone 154 (Scheme 62).¹⁸² Optimum reaction conditions comprised Pd(OAc)_2 and Cs_2CO_3 in toluene with molecular sieves at 80 °C. From among various ferrocene-base ligands, ferrocenyl phosphaneoxazoline L47 provided the best results. Decreasing the catalyst loading or lowering the reaction temperature resulted in a lower yield, but higher ee values were observed. The methodology worked well in the total synthesis of (–)-parvifoline.

Another approach toward α -aryl- α -fluorooxindoles 148 is an arylation of α -fluorooxindoles 149b (Scheme 63).¹⁸³ This reaction required more reactive aryl triflates as coupling partners. Under optimum reaction conditions (Cs_2CO_3 , Pd(dba)_2 /(*S*)-L29 catalyst in toluene), products were obtained in moderate to good yields (69–97%) and excellent

Scheme 63. Arylation of α -fluorooxindoles 149b

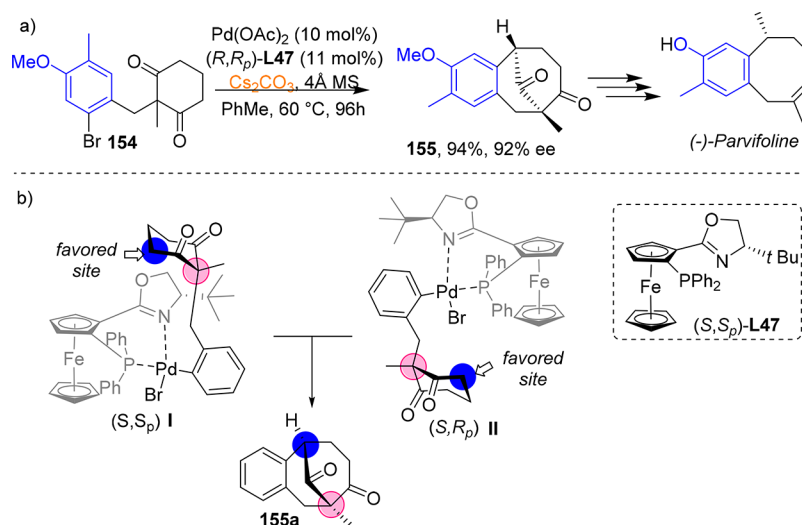
enantioselectivities (83–95% ee). *N*-Benzyl- and *N*-phenyl-protected 3-fluorooxindoles 149c could also be used.

■ COUPLINGS AND ALLYLIC ALKYLATIONS OF SILYL ENOL ETHERS AND KETENE ACETALS

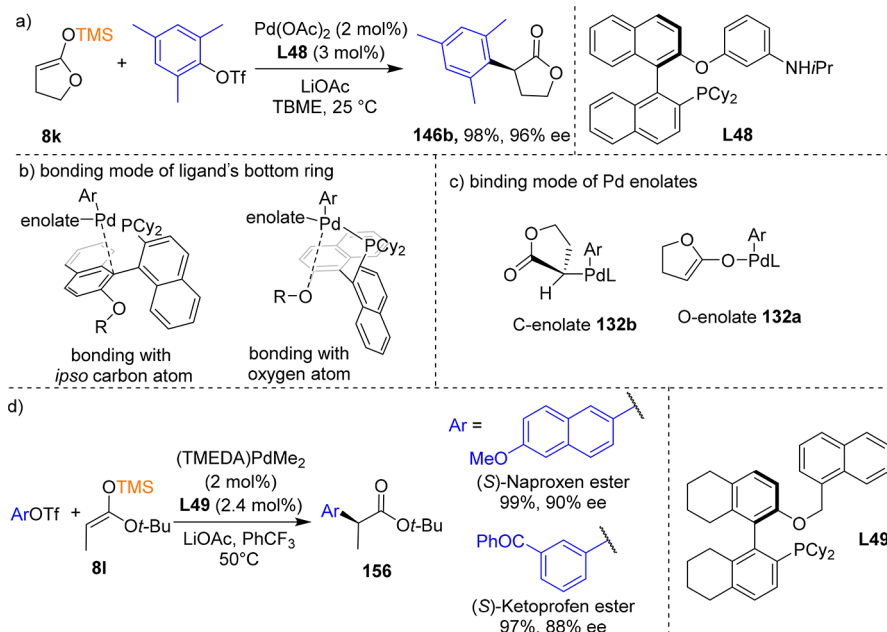
Zhou reported a Pd-catalyzed arylation of lactone 8j (Scheme 64a) promoted by weak arene hydrogen bonding (Scheme 64b).¹³⁸ Hydrogen atoms H1 and H8 of L48 can form hydrogen bonds with Pd-bound C enolates 132b (Scheme 64b,c). The aromatic amine groups can even form NH–O hydrogen bonds. Calculations have shown that *O*-enolates 132a were not present after transmetalation, which is contrary to popular belief. Since the palladium center is already coordinatively saturated by the lower ring of the biarylphosphine ligand, it was found that this is not the case. An unusually high barrier of $>15 \text{ kcal mol}^{-1}$ was found to be needed for an acetate-assisted enolate transfer, leading to *O* or *C* enolates 132. External attack of the OAc anion can favor enolate transfer with barriers of less than 10 kcal mol^{-1} . This methodology can be applied to a set of lactones 8k and worked well also on a gram scale. Lactones 8k can be opened and converted into amides, Weinreb amides, or ketones. To demonstrate the utility of the reaction, the authors synthesized two drug candidates—*o*-anisylkainic acid and (–)-preclamol.

Zhou also studied the cross-coupling of acyclic silyl ketene acetals 8l with aryl triflates (Scheme 64d).¹⁸⁴ LiOAc was chosen as the best activator for efficient coupling without racemization or double arylation, and the complex (tmeda)- PdMe_2 was the best Pd source. On the basis of the observed experimental data, the authors proposed a catalytic cycle. First,

Scheme 62. (a) Pd-Catalyzed Arylative Desymmetrization of 1,3-Diketone 154 and (b) Proposed Model Explaining the Stereochemical Outcome of the Reaction



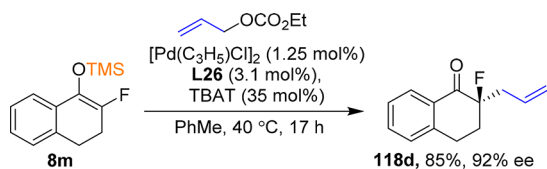
Scheme 64. (a) Pd-Catalyzed Arylation of Lactone **8k**, (b) Binding Mode of **L48** to Pd, (c) Binding Mode of Pd to Enolate, and (d) Cross-Coupling of Acyclic Silyl Ketene Acetals **8l** with Aryl Triflates



an oxidative addition forms the cationic species. Acetate anion binds to facilitate transmetalation of the enolate from silicon to palladium via an acetate-bridged structure. Ligand binding to Pd via phosphorus and the *ipso* carbon of the bottom ring of the ligand would presumably lead to the *S* product **156**. From the X-ray structure, it is apparent that the enolate would prefer to transfer to Pd from the left side. (*E*)- and (*Z*)-enolates give very different ee values of the products, indicating that a cyclic, closed transition state is present at the transmetalation step.

Enantioselective alkylation of fluorinated silyl enol ethers **8m** was also described using the Pd/**L26** catalyst (Scheme 65).¹⁸⁵ The Trost ligand and Josiphos ligands **L11** gave only traces or no product, (*R*)-**L21** provided the product in only 33% yield.

Scheme 65. Enantioselective Alkylation of Fluorinated Silyl Enol Ethers **8m**



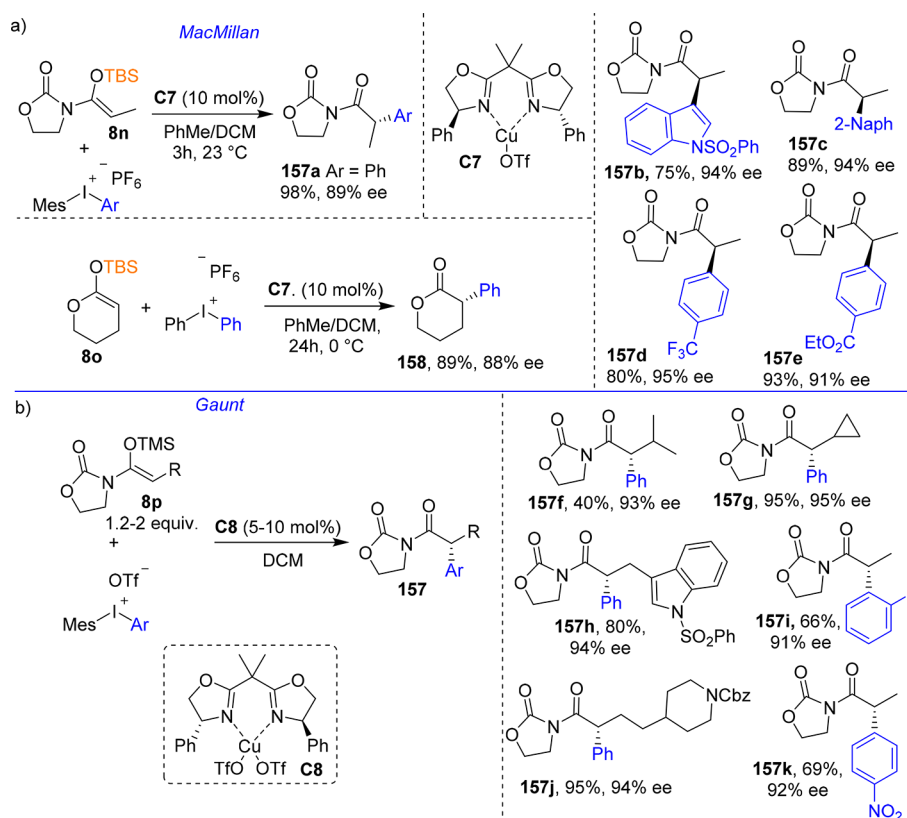
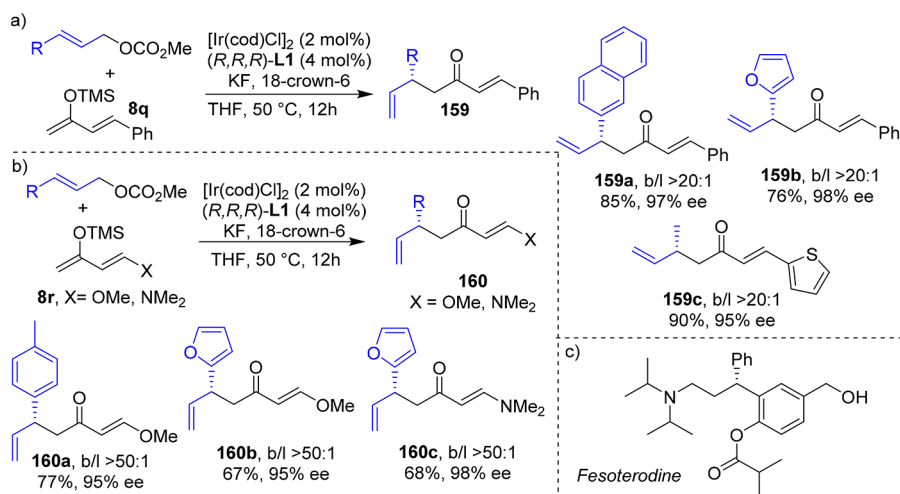
The MacMillan group reported an enantioselective α -arylation of carbonyls through their corresponding silyl ketene acetals **8n** or *N,O*-aminals (Scheme 66a).¹⁸⁶ The authors envisioned a mechanism involving an oxidative insertion of a ligand-copper(I) complex into a diaryliodonium salt. It would result in a highly electrophilic chiral copper(III) species, which would be captured by the silylated nucleophile. Reductive elimination and hydrolysis of the silyl would then yield the desired product **157** and regenerate the catalyst. Aryliodonium salts are very useful electrophiles, since they are nontoxic, readily accessible, and air and moisture stable. Although the silyl enol ethers **8** were reported to undergo α -arylation even without a metal catalyst, under these reaction conditions no product was formed without a catalyst.^{187,188} Phenyl-

substituted bisoxazoline catalyst complex **C7** gave the product of the coupling of *N,O*-aminal with the aryl iodonium salt in the highest yield and enantioselectivity, when OTf was used as the counterion. The use of the iodonium salt with the PF₆ counterion gave the product in a 98% yield and 89% ee. The protocol tolerates a wide range of functional groups, including esters and carbamates or silyl ketene acetals.

The Gaunt group reported a similar strategy in the same year (Scheme 66b).¹⁸⁹ They studied the reaction on *N*-acyloxazolidinones, which afforded the corresponding silyl ketene imide *Z* isomers **8p**. An advantage of the *N*-acyloxazolidinones in comparison to other carbonyl compounds is that they do not racemize as easily. Also, in this transformation, Cu(II)-bisoxazoline catalyst **C8** was effective for the enantioselective arylation. Instead of a biaryliodonium salt, a nontransferrable mesityl ligand was chosen, which is a useful strategy if a valuable aryl group is to be transferred.

Iridium-catalyzed allylic alkylation was first reported in 1997; since then the methodology has been widely used.^{190–192} Hartwig reported an iridium/phosphoramidite **L1** catalyzed enantioselective allylic substitution of silyl enolates **8q** derived from α,β -unsaturated ketones (Scheme 67a).¹⁹³ The key feature is the use of KF as an additive and 18-crown-6 to bind the alkali-metal ion. Allylic substitution with cinnamyl-, alkenyl-, and alkyl-substituted allylic carbonates gave the products in high yields and enantioselectivities. The synthetic utility of the method was demonstrated in the synthesis of a prostaglandin family member, TEI-9826.

A similar method was applied to Danishefsky and Rawal's diene **8r** (Scheme 67b).¹⁹⁴ This was a challenging reaction, since the product **159** contains a vinylogous ester, or amide, that could further react with the enol silane via an addition–elimination sequence. In this case, KF served as a base to promote the formation of the metallacyclic Ir catalyst, not as a Lewis base to bind and activate the enol silanes. The synthetic utility of the reaction was demonstrated in the synthesis of fesoterodine (Toviaz), an antimuscarinic drug.

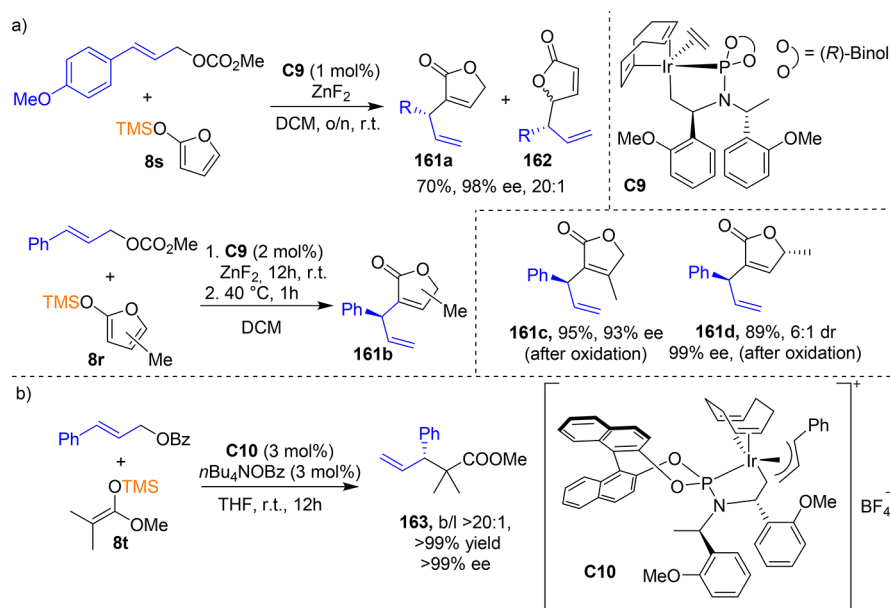
Scheme 66. (a) MacMillan's Enantioselective, Copper-Catalyzed α -Arylation and (b) Gaunt's Enantioselective α -Arylation of *N*-Acyl OxazolidinoneScheme 67. Ir/Phosphoramidite L1-Catalyzed Enantioselective Allylic Substitution of (a) Silyl Enolates **8q** and (b) Danishefsky and Rawal's Diene **8r**

3-Substituted butenolides **161** were obtained with high regioselectivity by an asymmetric allylic substitution between trimethylsiloxyfuran **8s** and cinnamyl carbonate catalyzed by iridium (Scheme 68a).¹⁹⁵ Optimized conditions comprised ZnF_2 iridacyclic phosphoramidite catalyst **C9** at room or slightly elevated temperature. Stoichiometric experiments revealed that the carbonate or benzoate generated in situ in the catalytic reaction activates the siloxyfuran. The iridium complex **C10** did not react with trimethylsiloxyfuran alone. The *R* configuration of the product can be explained by the nucleophilic attack *anti* to the isolated iridium fragment **C10**.

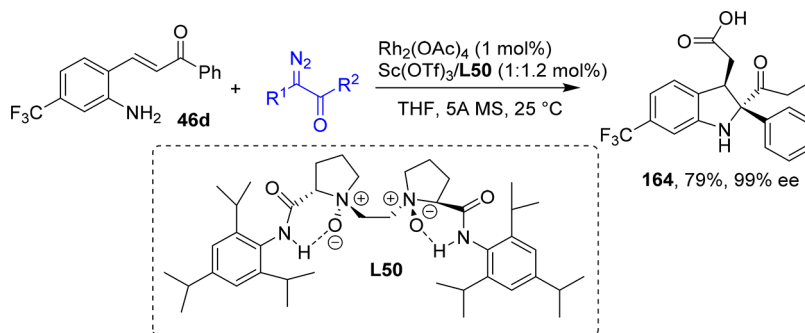
Other experiments suggested that the trimethylsiloxyfuran reacts directly with the Ir-allyl complex on activation by a carboxylate, not by the initial release of an allyl ester. The reaction gave excellent regio- and enantioselectivities with variously substituted butenolides **161**.

Iridium catalysis also worked well for enantioselective allylation of silyl ketene acetals **8t** (Scheme 68b).¹⁹⁶ The reaction of a cinnamyl methyl carbonate and ketene acetal **8t** was the model reaction. A catalytic amount of tetrabutylammonium acetate was added to activate the silyl ketene acetals **8**. More hindered leaving groups proved to suppress the

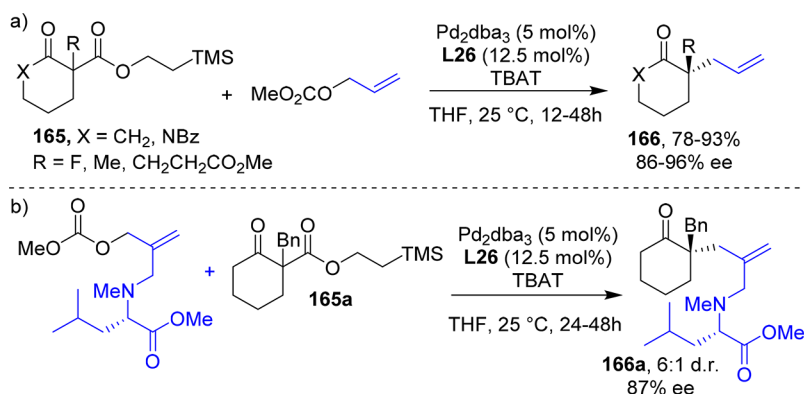
Scheme 68. (a) Ir-Catalyzed Asymmetric Allylic Substitution between Trimethylsiloxyfuran **8s** and Cinnamyl Carbonate and (b) Enantioselective Allylation of Silyl Ketene Acetals **8t**



Scheme 69. Liu and Feng's Enantioselective Synthesis of 2,2,3-Trisubstituted Indoline **164**



Scheme 70. Pd-Catalyzed AA of (Trimethylsilyl)ethyl β -Keto Esters **165**



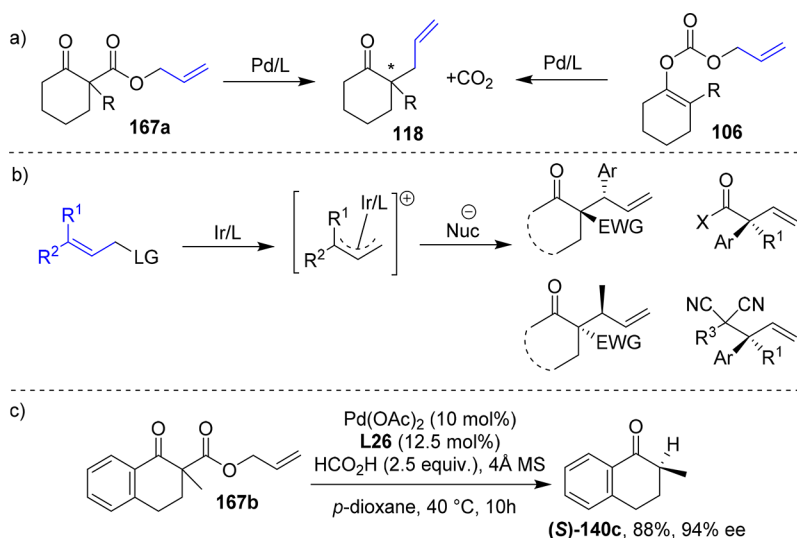
formation of an undesired side product; *tert*-butyl carbonate formed the product in 97% yield and 98% ee. The reaction of cinnamyl benzoate gave the product **163** in quantitative yield and >99% ee. The formation of undesired cinnamyl acetate was avoided using tetrabutylammonium benzoate.

■ UNUSUAL METHODS FOR ENOLATE FORMATION

Liu and Feng's enantioselective synthesis of 2,2,3-trisubstituted indoline compounds **164** was achieved by intramolecular

trapping of ammonium ylides generated from α -diazoketones by enones **46d** (Scheme 69). By employing $\text{Rh}_2(\text{OAc})_4$ /chiral N,N' -dioxide **L50** $\text{Sc}(\text{III})$ complex catalysis, the desired products **164** were obtained in high yield with enantioselectivity up to 99% ee and diastereoselectivity up to 19:1 (Scheme 69).¹⁹⁷ Balamurugan has reported annulations of the highly functionalized diazo-building block with indole under $\text{Sc}(\text{OTf})_3/\text{Rh}_2(\text{OAc})_4$ catalysis that proceeded through a Michael addition–cyclization sequence. The resulting tricyclic

Scheme 71. (a) Decarboxylative Tsuji–Trost Reaction, (b) Stoltz's Ir-Catalyzed AAs of Enolates, and (c) Enantioselective Decarboxylative Protonation of **167b**



compounds were obtained in good yield with excellent regioselectivity.¹⁹⁸

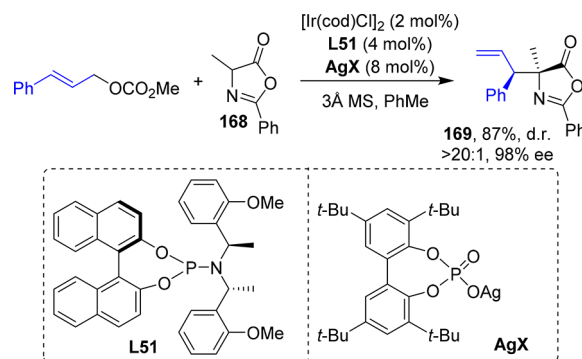
Stoltz introduced the(trimethylsilyl)ethyl ester protected enolates **165**.¹⁹⁹ (Trimethylsilyl)ethyl β -keto esters **165** undergo enantioselective AA after deprotection by a fluoride source. The advantage of this methodology is that no base is needed; therefore more complex and base sensitive structures can be prepared (Scheme 70).

Decarboxylative allylation and benzylation reactions represent a rapidly developing new field of enolate chemistry.^{200,201} Allyl enol carbonates **106** or β -ketoallyl esters **167a** are used to prepare α -quaternary cycloalkanones **118** under Pd catalysis (Scheme 71a). The allyl carbonate moiety is simultaneously applied for generating the enolate in situ and as an allylating agent.

The Stoltz group also widely contributed to the development of Ir-catalyzed AA reactions of enolates (Scheme 71b).²⁰² They described Tsuji–Trost allylation using allyl enol carbonates **167** and silyl enol ethers **8**,²⁰³ as well as the construction of asymmetric α -quaternary cycloalkanones **118**.^{200,201} Stoltz has also reported enantioselective decarboxylative protonation to yield tertiary stereogenic centers (Scheme 71c).²⁰⁴ The reaction possibly proceeds through the chiral Pd enolate intermediate **132**, which is consistent with the observed reactivity. The enolate then reacts with an electrophile, in this case a proton, to yield the tertiary stereocenter.

Azlacones **168** are popular building blocks in organic synthesis.²⁰⁵ They can be used as enolate equivalents in transition-metal-catalyzed transformations. Asymmetric allylic substitutions catalyzed by iridacyclic phosphoramidite complexes usually occur with low diastereoselectivity, albeit the diastereomers usually had high enantiomeric purity. Unsymmetrical allyl carbamates were used for enantioselective allylations of azlacones **168**. Allylic allylation of azlacones **168** can be catalyzed by palladium²⁰⁶ or molybdenum, as was described by Trost.²⁰⁷ Iridium-catalyzed, highly diastereo- and enantioselective allylation of azlacones catalyzed by a metallacyclic iridium complex and achiral phosphate was introduced by Hartwig (Scheme 72).²⁰⁸ The silver salt **AgX** was used to generate the metallacyclic iridium catalyst in situ.

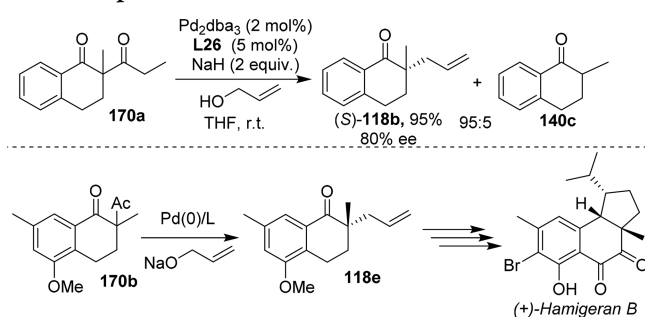
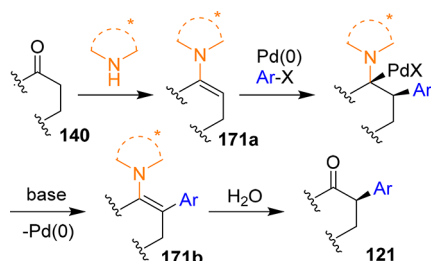
Scheme 72. Ir-Catalyzed Allylation of Azlacone **168**



The counteranion deprotonates the pronucleophile and modulates the degree of stereocontrol at the new chiral center of the nucleophile. The counterion for control of the diastereoselectivity is thought to be introduced into the catalytic cycle by generating the metallacyclic iridium catalyst in situ from silver salts. A cation-controlled diastereoselectivity was described for the allylation of 5H-oxazol-4-ones and 5H-thiazol-4-ones by employing the corresponding zinc (**2**) or magnesium (**36**) enolates.²⁰⁹

Although decarboxylative allylation represents an attractive approach since it does not require basic conditions to generate the enolate, the coupling of the allylic electrophile is necessary before the reaction. Careful control of the O vs C alkylation is needed and may not be achieved for all kinds of substrates. Tunge and co-workers introduced direct decarboxylative allylation of ketone pronucleophiles **170** (Scheme 73).²¹⁰ Subsequent retro-Claisen cleavage resulted in the formation of the active nucleophile and simultaneously activated the electrophile toward Pd-catalyzed coupling. Biologically active (+)-hamigeran was prepared using this methodology.

Enamines **171** are similar in reactivity to enolates, and their metal-catalyzed α -arylation was introduced (Scheme 74).²¹¹ A combination of MacMillan's organocatalysis to prepare chiral enamines **171** and subsequent metal catalysis to form the new C–C_{Ar} bond was used. Since this methodology is not directly

Scheme 73. Deacylative Allylation of Ketone Pronucleophiles**Scheme 74. Combination of Enamine and Pd Catalysis**

related to our topic, and there was a recent review about dual amino and metal catalysis, we will not go into detail here.²¹¹

Another topic related to this review, which has been flourishing recently, is synergistic catalysis, which combines the action of an organocatalyst with a transition-metal catalyst.^{212–215}

Hartwig recently reported a stereodivergent allylation of esters **172**.²¹⁶ The Lewis basic chiral amine (*S*)-BTM was used to displace the phenoxide anion of the ester and generate an ammonium enolate. Iridium catalyst **C10** was then used for the AA. On the basis of the stereochemistry of the LB and Ir ligand all four possible stereoisomers are accessible. Hartwig also reported a synergistic iridium and copper catalysis using the chiral ferrocene-based bisphosphane ligand **L11** (Scheme 75).²¹⁷ After deprotonation, a theCu(I) enolate **100** is obtained, with a well-defined geometry thanks to the chelation. The enolate then reacts with the electrophile catalyzed by a chiral iridium complex. By using a different combination of chirality on Cu and Ir, all four possible stereoisomers are accessible.

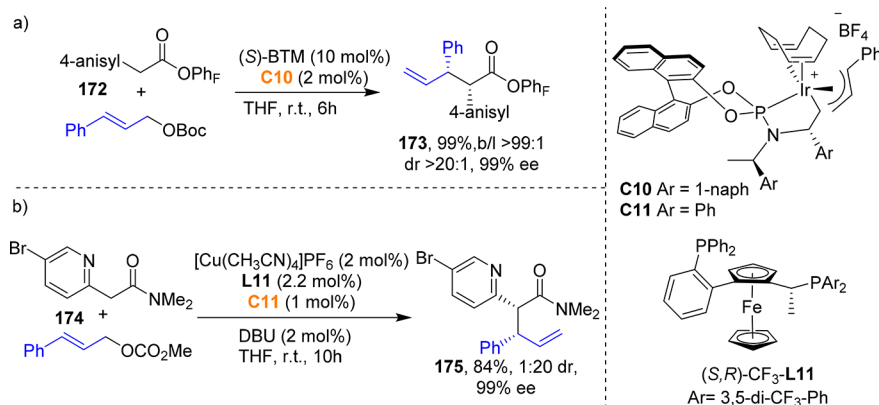
Zhao and co-workers developed Ti(IV)-assisted Pd-catalyzed divergent synthesis of [5,5]- and [6,5]-spiro-heterocycles **176** and **177**. Using Mg(*t*-BuO)₂ [5,5]-spirocycles **176** can be formed in quantitative yield as a single diastereomer from carbonates **178**. However, by employing the Ti(*i*-PrO)₄ instead, the selectivity of the reaction is inverted completely in favor of [6,5]-spiro-heterocycles **177** (99% yield, >20:1 dr), which can be explained by the reversed polarity of **178** (Scheme 76).²¹⁸

CONCLUSIONS

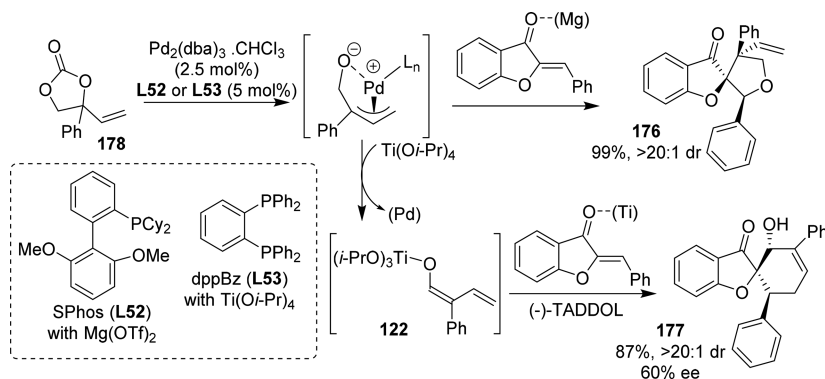
The use of enolates in asymmetric synthesis has strong roots in the first tandem zinc enolate **2** reactions, which allowed total stereocontrol of the reaction and afforded highly desired compounds in the total synthesis of natural products with high enantiomeric purities. The methodology has since developed into a universal method for functionalization of a variety of other transition-metal enolates, thus allowing fine-tuning of reaction conditions to obtain even more sensitive or challenging substrates. Two main strategies for obtaining reactive metal enolates are deprotonation of the sufficiently acidic compound with suitable metal bases and transition-metal-catalyzed conjugate additions of organometallic reagents. Asymmetric conjugate additions of dialkylzinc, Grignard, trialkylaluminum, organoboron, or organozirconium reagents were utilized to form the corresponding metal enolates, with the added benefit of another stereogenic center already created. These chiral enolates have been employed in many trapping reactions such as aldol, Mannich, or alkylation reactions, generating a range of structurally and stereochemically complex compounds, including natural products. An interesting strategy is also CuH-catalyzed conjugate reduction followed by functionalization of either boron or silicon enolates. Products obtained by this strategy have new C_{sp}³–C_{sp}³ bonds.

The formation of a new C_{sp}³–C_{Ar} bond is achieved by metal enolates generated by deprotonation and weakly reactive electrophiles such as aryl and alkenyl halides and pseudohalides. Pd or Ni catalysis is required to promote the reaction. S_N2-type electrophiles were used for allylic alkylation reactions catalyzed by chiral Ir and Pd complexes.

Among challenges that are still unresolved is building a new C_{sp}³–C_{Ar} stereogenic center by a one-pot enantioselective conjugate addition with a TM-catalyzed cross-coupling reaction. Another potentially useful, but for now unreachable,

Scheme 75. Stereodivergent Allylation of Esters **172 and Amides **174** Using a Combination of Lewis Base and Ir Catalysis**

Scheme 76. Aldol Reaction of Ti Enolates 122



strategy might be finding a more general coupling methodology for the base-generated enolates, or perhaps a universal ligand, since the reactions are often very sensitive to small changes in the reaction conditions or in the ligand structure which can lead to the absence of product formation. A methodology for building stereogenic centers with non-functionalized sp^3 -hybridized electrophiles in the cross-coupling step is also missing at this point.

With this review, we aimed at providing the catalysis community with an overview of the recent advances in generation and functionalization of chiral metal enolates and their functionalization via direct or transition-metal-catalyzed processes. We hope that, in this way, we can help stimulate the further development of this important facet of synthetic organic chemistry.

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Notes

The authors declare no competing financial interest.

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